

The Effect of Conditioned Mergers on Underlying Representations

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1 Introduction:

Conditioned Mergers and Representational Ambiguity

A conditioned merger is a type of phonological rule whereby two underlyingly distinct phonemes share an identical phonetic realization in some proper subset of possible phonological environments. After such a merger has taken hold in a given language, that language now contains instances of a merged phone that are ambiguous between two underlying phonemes. This presents a problem from a language acquisition perspective; by what means does a child in acquisition decide on which underlying representation the ambiguous phone should be assigned?

Oftentimes it may be possible to identify some general rule R by which the child selects one of two possible underlying representations rather than the other by default. However, in some cases, it may not be obvious how to identify such a rule or what UR individuals use to represent a merged phone. An interesting example of this arises in the case of intervocalic flapping in North American English. Consider the homophonous word-pair *latter* and *ladder*, both pronounced $/læɾə/$. The flapping literature largely takes for granted that adults represent $/ɾ/$ instances faithfully to their historical/orthographic value, i.e. $/t/$ and $/d/$ for *latter* and *ladder* respectively. However, given that the invariability of the merged flap pronunciation in a number of phonemes (including *latter* and *ladder*, which do not generally surface with unflapped realizations), it is unclear why children should assign different URs for two different flaps that have the same phonological patterning in the stimulus. Furthermore, given that the etymological value of a non-alternating instance of $/ɾ/$ is arbitrary from a synchronic perspective, it is unclear how it should even be possible for a child in acquisition to faithfully differentiate between $/t/$ - and $/d/$ -flaps without referring to metalinguistic factors such as spelling. An exploration of ambiguities of this type in the representation of merged phones such as flaps constitutes the first part of this work, given in Section 2.

Whether or not there exists an identifiable default rule R for UR assignment to ambiguous merged phones, the ambiguity of merged phones may also be resolved directly by disambiguating evidence, namely of the following types:

1. Environmental alternations – the merger may be bleb in certain environments, resulting in unmerged alternants of the phones in question e.g. in French, most word-final consonants historically underwent deletion (i.e. merger with $\emptyset$), but the bleeding of this rule by re-syllabification with a following vowel-initial word (called *liaison*) resulted in $[C \sim \emptyset]$ alternations that allowed for identification of ambiguous morpheme-final $[\emptyset]$ as having an underlying consonant, e.g. $/z/$ in *les* ‘the’, normally realized $[le]$ except pre-vocally, where the realization is $[lez]$
2. Variable application of the merger – unmerged tokens of a variable merger allow for accurate identification of URs, e.g. medial schwa deletion in English applies variably, resulting in some ambiguously $/ə/$ -less realizations which can be disambiguated through tokens without $/ə/$ -deletion, e.g. *elaborate* (adj.) $[\ə'læbɪrət]$ disambiguated by $[\ə'læbɪrət]$
3. Incomplete neutralization – a phonetic trace of an underlying phonological distinction prevents two phonemes from becoming fully phonetically identical in a conditionally merging environment. There are two subtypes:
 - (a) the ‘merged’ phonetic realizations of the two phonemes themselves are not fully identical, e.g. it has been (inconsistently) reported that $/d/$ -flaps are slightly longer than $/t/$ -flaps. The fact that such distinctions are typically not perceptible calls into question the notion that they can be used as a cue of UR.
 - (b) allophony in adjacent segments that is counterbled by the merger in question may cue the UR of a merged phone, e.g. in many varieties of North American English, $/aɪ/$ has a raised allophone $/ɪ/$ before underlyingly voiceless segments, allowing differentiation of otherwise homophonous *writer* $/ˈaɪtɪə/$ $[ˈɪaɪtɪə]$ and *rider* $/ˈaɪdɪə/$ $[ˈɪaɪdɪə]$

It is easy enough to see that these three types of disambiguating evidence is an exhaustive list of the types of evidence available from the linguistic stimulus itself: either evidence may come from internal differences

in the merged phonemes themselves, which may be due to environmentally conditioned differences in the realization of those phonemes, environmentally independent yet variable differences in the realization of those phonemes, or consistent internal differences in the realization of those phonemes (corresponding to 1, 2, and 3a, respectively), or evidence may come from external differences in the environment that reflect the underlying differences of the merged phonemes (corresponding to 3b). However, even if these are the only strictly linguistic means of ascertaining the UR or a flap, there remains the (controversial) possibility that metalinguistic factors like spelling may play a role in UR assignment. This possibility will be explored in Section 2 as well.

Even in cases where disambiguating evidence of the types enumerated above exists, we often observe erosion of the historical phonological opposition between the two merging phonemes in the environment of the merger. In the second section of this work, I consider two liaison systems that have demonstrated such erosion, namely /ɹ/-liaison in non-rhotic English varieties and French liaison. Both systems historically arose under very similar phonological circumstances, i.e. the bleeding of word-final consonant deletion (i.e. merger with $\emptyset$) by re-syllabification with a following vowel-initial morpheme, yielding disambiguating evidence through resulting environmental alternations. However, the two systems have followed markedly different paths of development. In English, /ɹ/-liaison has expanded its environment from morphemes with historic /ɹ/ (which have so-called linking /ɹ/) to morphemes with no etymological /ɹ/ (with so-called intrusive /ɹ/). Contrast this with French liaison, where the set of morphemes in which liaison can take place has dramatically reduced in size. Although it purportedly could occur for any word-final consonant historically, it now only occurs in a very restricted set of morphemes, specifically function words, inflectional suffixes, and a small group of adjectives and adverbs, as well as in a few set frozen expressions.

In the second part of this work, Section 3, I argue systematic erosion of this type is an expected result of the limitations of disambiguating evidence of the type described above in the assignment of URs to ambiguous phones after mergers. Additionally, I evaluate crucial differences in the English and French liaison systems that may help elucidate the reasons for the two systems' apparently opposite paths of development.

2 Merged Phones without Disambiguating Evidence: Morpheme-Internal Intervocalic Flaps

In cases where an instance of a merged phone truly has no disambiguating evidence for its underlying representation, the question of what underlying phoneme individuals assign to this phone becomes problematic. One possible solution comes from the principle of Invariant Transparency, given in (1) as formulated by Ringe & Eska (2013):

- (1) 'Native learners in the developmental window for NLA do not posit abstract forms if there is no alternation to account for; instead they project surface segments, defined by the languages system of distinctive features, into underlying forms'

Thus, children may simply assign a UR corresponding most closely to that the surface pronunciation of that phone. However, conditioned mergers almost always involve the alternation of at least one of the two merging phonemes with that phone; the fact that the phonemes remain phonetically distinct outside the merging environment implies that at least one of those phonemes has a realization that is distinct from its realization inside the merging environment (unless the phonological effects of the conditioned merger involve one of the merging phonemes no longer occurring in the environment of the merger and no environmental alternations that could bleed the merger exist). In cases where one of the phonemes has the same realization inside and outside the merging environment, we can likely assume that the UR assigned prior to a child's recognition of any alternations involving that phone persists unless disambiguating evidence occurs. For example, before a child learns about a nasal to velar assimilation rule, whereby /n/ may become [ŋ] before a velar consonant, *sink* [sɪŋk] would be assigned /sɪŋk/. Even once the child identifies the nasal assimilation rule from words like *pancake* ['pæŋkeɪk], they will not re-assign *sink* the UR /sɪnk/, even though this UR predicts the same surface realization and is hence just as phonologically consistent as /sɪŋk/.

However, the situation becomes more complicated when the merged phone is not the same as the phonetic realization of either of the merging phonemes outside the merging environment, namely because maintaining the same UR after the recognition of the allophonic relationship between this phone and the two merging phonemes would require an additional, unnecessary segment in the phonemic inventory of that language. For reasons of parsimony, such analyses involving more phonemes than necessary are generally dispreferred.

An instance of this latter scenario occurs in the case of intervocalic flapping in a number of English dialects, whereby /t/ and /d/ have the merged phone [ɾ]. For reasons of parsimony (or perhaps simply convention), it is generally not assumed that [ɾ] ever has the underlying value /ɾ/, and the literature largely assumes that the underlying representation of a flap in a given morpheme is faithful to the historic (and correspondingly orthographic) value of that flap. However, this requires a non-trivial assumption that individuals have a reliable strategy by which to faithfully infer URs for flaps that are consistent with their historic value. In some cases, it can be shown that no such reliable strategy exists without referring to metalinguistic inputs like orthography. Hence, intervocalic flapping and other conditioned mergers of this sort are fraught with ambiguity regarding linguistic cues for underlying representations. The next sections of this work are concerned with an exploration of how ambiguity of this type is resolved on the basis of findings from past research as well as new research using flapping data from the Philadelphia Neighborhood Corpus.

2.1 Phonology of Intervocalic Flapping

Intervocalic flapping is a conditioned merger of the coronal stops /t/ and /d/ into a voiced alveolar flap [ɾ] that occurs in a variety of English dialects, most notably North American and Australian English.

The rule-based definition of intervocalic flapping may be formulated as follows (Bermúdez-Otero 2004):

$$(2) \quad /t,d/ \rightarrow [ɾ] / \left\{ \begin{array}{l} \text{I_V} \\ \text{V_V} \end{array} \right. \text{ under certain prosodic conditions.}$$

The exact prosodic conditions of rule (2) have a complex relation with stress, syllable structure, and intonational phrasing, especially at word boundaries. Word-finally, flapping is somewhat variable, though it is generally licensed for any morpheme-final flap given a following vowel-initial word in the same prosodic phrase regardless of the relative stress of the adjacent syllables, e.g. *get over it* [gɛɾ 'oʊvəɪt]. Word internally, flapping generally occurs when the following vowel is unstressed, however it can also occur pre-tonically given that the coronal stop is morpheme-final, e.g. *whatever* [wəɾ'evəɪ]. These environments can be unified under a cyclic analysis where flapping is licensed if the coronal stop occurs in coda position intervocalically at some point in the derivation (independent of stress). Although flapping is variable word-finally, word-medially it is more consistent, though some speakers show significant variability particularly between unstressed vowels in certain words, e.g. *reality* [ɹi'ælɹɪ ~ ɹi'ælɹɪ].

One environment where flapping is thought to occur categorically is intervocalically, morpheme-internally after a stressed syllable before an unstressed syllable, e.g. *better* [bɛɾə]. Flapping in this environment produces a bountiful number of homophones in affected English dialects, as for example in (3):

	Word	Historic UR	Current Surface Realization
(3)	latter	/ˈlætər/	[ˈlæɾə]
	ladder	/ˈlædər/	[ˈlæɾə]
	bleating	/ˈblɪtɪŋ/	[ˈblɪɾɪŋ]
	bleeding	/ˈblɪdɪŋ/	[ˈblɪɾɪŋ]

An exception to categorical flapping in the /V̌_V/ environment worth noting occurs before syllabic /ŋ/, where it is bled by /t/-glottalization, e.g. *button* /ˈbʊtən/ [ˈbʌʔŋ]. This bleeding effect also may occur with coronal variants of the *-ing* suffix, e.g. *gettin'* /ˈɡetən/ [ˈɡɛʔŋ]

2.2 Disambiguating Evidence for Flap URs (and Lack Thereof)

As discussed in the introduction, the three types of (strictly non-metalinguistic) disambiguating evidence for underlying representations of conditionally merged phonemes come from environmental alternations, variability, and incomplete neutralization. In the next, three subsections, we will consider the availability of evidence for flap URs falling into each of these categories in turn. It will be shown that while learners of English may be able to utilize disambiguating evidence of this sort, there exists a subset of flaps for which sufficient disambiguating evidence is unlikely to occur, namely morpheme-internal flaps in the / $\acute{V}$ _V/ environment (excluding environments before a syllabic nasal).

2.2.1 Environmental Alternations and Flapping

Environmental alternations between ambiguous flaps and unambiguous faithful stops /t,d/ occur most frequently morpheme-finally. Such alternations may occur either word-internally, e.g. *hitter* [hɪrə] vs. *hit* [hit], or word-finally, e.g. *hit a run* vs. [hɪr ə ɹʌn] vs. *hit* [hit]. Generally, morpheme-final stops that alternate between flapping and non-flapping environments occur much more frequently in non-flapping environments. Thus, it appears reasonable to assume that children in acquisition have sufficient evidence for the underlying representations of morpheme-final flaps, since flaps are a secondary rather than primary realization of coronal stops in this case.

Additionally, a much more limited number of morphemes show alternations between flaps and faithful stops due to stress alternations, e.g. *atom* [ˈæɾəm] vs. *atomic* [əˈtʰəmɪk]. Such cases are mostly limited to Latinate vocabulary, much of which is more learned. The overwhelming majority of morpheme-internal flaps do not alternate with faithful stops as the result of stress alternations of this sort. Hence, alternations of this sort provide an at best marginal source of disambiguating evidence for morpheme-internal flap URs.

2.2.2 Variability and Flapping

As already indicated, flapping is variable word-finally. Since word-final environments are already highly susceptible to environmental alternations, this variability serves to reinforce the disambiguating evidence that comes from occurrences morpheme-final coronal stops in non-flapping environments.

However, word-internally, particularly in the / $\acute{V}$ _V/, non-pre-nasal environment, it does not seem that variability can be used as a disambiguating cue for UR assignment, since flapping applies categorically in these environments. Zue & Laferriere (1979) found that 99% of underlying stops in this environment were flapped by educated American subjects in a wordlist experiment. An analysis orthographic /t/ tokens also in the / $\acute{V}$ _V/ environment (again excluding tokens before syllabic nasals) produced by speakers of the Northern and North Midland dialects of American English in the SWITCHBOARD corpus by Patterson & Connine (2001) found a slightly lower, but nevertheless quite high flapping rate of 93.9%. If flaps are seen as the “correct target pronunciation, the difference between the two studies might be accounted for by the fact that data of the former study are more controlled than those of the latter. However, it is also plausible that speaker in the SWITCHBOARD corpus, speaking over the phone, were more likely to hyperarticulate target flaps as faithful stops. At any rate, the relevant take-away for our purposes is that intervocalic flapping is near categorical.

Patterson & Connine (2001) also found two relevant conditioning factors in the rate of flapping. Firstly, infrequent words, defined as words with lexical frequency of less than 60 per million, are much more likely to be pronounced with a non-flapped realization (either /t/ or /ʔ/) than frequent words (23.9% of 355 tokens vs. 4.1% of 3,758 tokens). This may back the interpretation that the comparatively higher rate of non-flapping is the result of hyperarticulated pronunciations, since infrequent words are known to have longer durations and are more likely to be hyperarticulated.

Secondly, and perhaps more interestingly for our purposes, non-flapped pronunciations were more common morpheme-finally (23.9% of 253 morpheme-final tokens vs. 5.6% of 3840 morpheme-internal tokens). Again, since morpheme-final flaps already have disambiguating evidence from environmental alternations, this means

that much of the already highly limited amount of variable non-flapping word-internally redundantly goes to help disambiguate flaps that are already likely to have disambiguating evidence. Thus, variable non-flapping is unlikely to provide an adequate disambiguating cue for the underlying voicing of morpheme-medial flaps.

2.2.3 Incomplete Neutralization and Flapping

Incomplete neutralization, alluded to in the introduction, occurs when a phonetic trace of an underlying phonological distinction prevents two phonemes from becoming fully phonetically identical in a conditionally merging environment. Generally, this occurs due to allophony of an adjacent segment that is differentially conditioned in a counter-bleeding order by the features of the phonemes implicated in a conditioned merger. The canonical example of German coda devoicing, where voiced obstruents are merged to voiceless obstruents in the syllable coda, e.g. *Rat* /ʁa:t/ “advice” and *Rad* /ʁa:d/ “wheel”, both pronounced [ʁa:t] (Braver, 2014). Although, these words are judged by native speakers to be homophonous, phonetic analysis reveals that vowels before underlyingly voiced coda obstruents are on average 15 ms longer than before their underlyingly voiceless counterparts (Port & O’Dell, 1985).

Incomplete neutralization in the German coda devoicing case is related to a cross-linguistically frequent phenomenon called *pre-fortis clipping* (PFC), whereby vowels are reduced in duration when followed by a voiceless segment (Messum 2008). PFC occurs in English as well as in German (and purportedly in most of the world’s languages), and a growing number of studies have reported that PFC results in incomplete neutralization of English /t/ and /d/-flaps, similarly to the classic German example, with differences in average vowel duration before /t/ and /d/-flaps of 5.69 ms in a wug task (Braver 2014), 7 ms in a wordlist task (Zue & Laferriere 1979), and 16 ms in minimal pairs extracted from a large corpus (Patterson & Connine 2001), even though these studies fail to find any difference in the production of /t/-flaps and /d/-flaps themselves (e.g. in flap duration).

The fact that vowel duration differences resulting from PFC behave opaquely with respect to the underlying voicing of the following segment has led many authors (Fox & Terbeek 1977, Zue & Laferriere 1979, Bermúdez-Otero 2004) to conclude that PFC is a phonological rule in American English, rather than merely a phonetic effect, and that this rule occurs before application of the flapping rule in a counter-bleeding relation. There are a couple of potential issues here. Firstly, even though vowel durations are shorter before /t/-flaps than /d/-flaps as predicted by a phonological PFC rule, all pre-flap vowel durations, even those before /d/-flaps, fall into the range of short vowels (Fox & Terbeek 1977). Fox & Terbeek (1977) argue that there is an additional pre-flap shortening rule that shortens vowels before flaps to some fraction of their original length, though they note that “such graduated rules do not occur in current generative phonologies (p. 33). Bermúdez-Otero (2004) avoids this problem by positing a “gradient, non-neutralizing phonetic process of durational adjustment reduces, but does not eliminate, the categorical length difference between clipped and unclipped vowels before flaps (p. 5), though he does not offer any articulatory motivations for such a phonetic process.

Secondly, positing a phonological PFC rule would contradict claims in the literature that pre-fortis clipping is a strictly phonetic coarticulatory effect. For instance, Messum (2008) argues that PFC results from the fact that voiceless obstruents involve greater articulatory effort than voiced obstruents, necessitating that effort be reallocated towards the production of voiceless obstruents at the expense of adjacent segments. PFC has been observed to occur in nearly all languages investigated, suggesting it is a phonetic universal. Even in one rare exception of a language without PFC, Swedish, children at 24 months have been observed to demonstrate PFC before suppressing it by 30 months, presumably to account for phonemic vowel length in their language, strongly suggesting a phonetic rather than phonological phenomenon since Swedish children never hear PFC in the target language and therefore could not plausibly learn it as a phonological rule on the basis of the stimulus (Buder and Stoel-Gammon 2002). Of course, it is not necessarily the case that these accounts of PFC as either phonetic or phonological are wholly incompatible with one another; PFC could be a natural phonetic phenomenon that has undergone phonologization in American English. After all, phonologization of phonetically motivated contrasts has long been considered a major potential source of sound change.

While we cannot offer a definitive solution to these problems at present, for our purposes the important

thing is that studies have demonstrated that the underlying voicing of flaps affects the application of PFC, resulting in longer vowels before /d/-flaps than /t/-flaps. Accepting this, the question becomes whether pre-fortis clipping provides sufficient disambiguating evidence for the underlying voicing of flaps. This seems highly unlikely for a couple of reasons.

Firstly, adult native American English speakers are unable to perceptually distinguish between /t/- and /d/-flaps, performing at essentially chance levels. Braver (2014) finds that adult speakers are unable to perceptually distinguish /t/-flaps and /d/-flaps, despite deliberately using tokens with more extreme duration differences in his experimental design, with 48.78% of /d/-flaps and 49.49% of /t/-flaps correctly identified. Given, this fact, it seems unlikely that children would be much better at this task.

Secondly, even if we suppose that children in acquisition are more attuned to fine phonetic differences than adults are (which is often true), it does not seem plausible that individuals of any age are capable of reliably hearing and making use of the reported mean duration differences of 6 to 20 ms (i.e. up to a 50th of a second). This seems particularly doubtful given the high variability of speech rate and the fact that vowel durations before /t/- and /d/-flaps have heavily overlapping distributions.

Thirdly, the fact that vowel durations even before /d/-flaps fall into the range of pre-voiceless vowels, it seems probable that children’s use of vowel durations, if they do indeed make use of such differences, would lead them to assign /t/ to all or most ambiguous flaps, whether spelt with *t* or *d*, since this is more consistent with phonological rules that are more easily and readily discernable to them.

Investigators have also attempted to identify minor phonetic differences between /d/-flaps and /t/-flaps themselves, particularly in flap duration. However, studies have generally found no significant difference. One exception is Herd et al. (2010), which found a 0.9 ms difference. Although miniscule, this difference reached levels of significance. However, the authors attribute the significance of this difference to a large sample size, and assert that flap duration is not a cue for underlying voicing.

Even though pre-fortis clipping and flap duration are unlikely to provide a sufficient cue for the assignment of flap URs, there exists at least one salient allophonic rule sensitive to the underlying voicing of flaps which is well-known to apply in a counter-bleeding order with flapping and can therefore act as a cue for flap URs, namely Canadian raising. Canadian raising occurs in a variety of North American English dialects and refers to the allophonic raising of the diphthong /aɪ/ to /ʌɪ/ before voiceless consonants¹, e.g. *ride* /ɪaɪd/ [ɪaɪd] but *write* /ɪaɪt/ [ʌɪt]. Since Canadian raising occurs in a counterbleeding order with flapping, we also observe near-minimal pairs with flaps where only the /aɪ/ allophone cues an underlying phonological difference, e.g. *rider* /ɪaɪdər/ [ɪaɪdər] and *writer* /ɪaɪtər/ [ʌɪtər].

There are a couple of shortcomings to the use of Canadian raising as a cue for flap URs. Firstly, it applies only in an extremely restricted set of cases, namely only after the vowel /aɪ/ (and for some speakers /aʊ/). Secondly, for at least some dialects, it can be shown that Canadian raising developed after the rise of flapping, so even in cases where Canadian raising could provide evidence of flap URs, in such dialects Canadian raising as an underlying voicing cue could not have provided continual transmission of flap URs over generations since the inception of the merger of /t/ and /d/ through flapping.

Additionally, the interaction of flapping and Canadian raising reveals some strange behavior. A tendency for words with orthographic /d/-flaps to surface unexpectedly with raised /aɪ/ allophones, e.g. *spider* [spaɪər], *cider* [saɪər], etc., has been observed in a variety of dialects, including Philadelphia (Fruehwald 2008), the Inland North (Dailey-OCain 1997), North Central (Vance 1987). Purported “lexical exceptions” to Canadian raising of this type also occur in some other words for some speakers, e.g. *cyber* [saɪbər], *tiger* [taɪgər], but exceptions with orthographic /d/ are the most frequent and also the most consistent across speakers.² It has been noted that by researchers such as Fruehwald that exceptional /aɪ/-raising before /d/-flaps may arise though association with regular raising before /t/-flaps. One possibility that has not been directly raised (to my knowledge) is that such raising “exceptions” are actually exceptions at all, but rather that some

¹Canadian raising also involves alternation of /aʊ/ with /ʌɪ/, but this change occurs in a much more restricted geographical area, namely Canada and some parts of New England and the Upper Midwest, whereas raising of /aɪ/ includes broader swathes of the Midwest and New England as well as New York down through Philadelphia

²I come from Upstate NY, and I pronounce /aɪ/ with a raised allophone quite generally before all instances of morpheme-internal flaps.

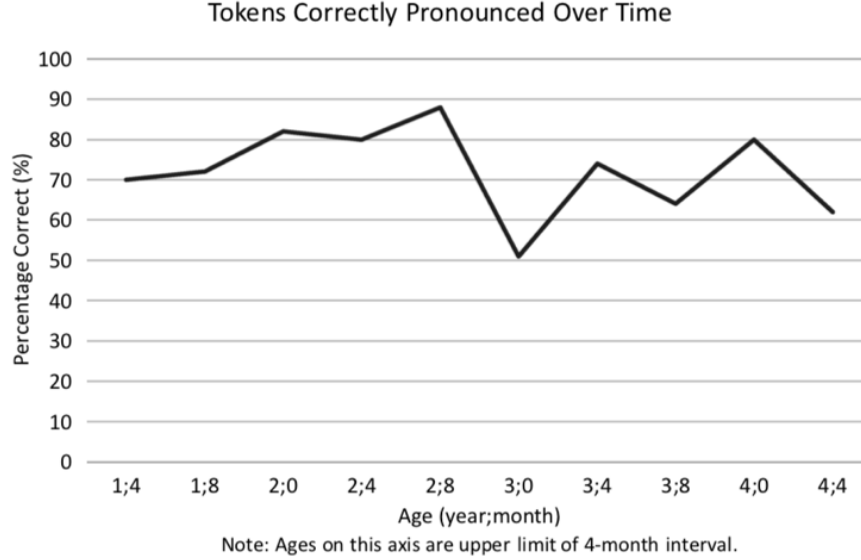


Figure 1: Correct flap productions in obligatory contexts by one child (CHILDES: Providence/Lily) (Richter 2018 p. 666)

historic /d/-flaps have been reanalyzed as /t/-flaps, perhaps as a result of the difficulty of ascertaining the underlying voicing of morpheme-internal flaps, and raising therefore applies regularly. In any case, the fact that raising exhibits inconsistent behavior in its interaction with flapping suggests that it may not be as robust a cue for flap URs as we might otherwise expect.

Synthesizing all of this background and analysis, we see that there is no clear cue for the underlying voicing of flaps morpheme-internal in the $/\check{V}_V/$ environment, excluding environments before syllabic nasals (and potentially after /aɪ/ and /aʊ/ in dialects with Canadian raising of these phonemes).

2.3 Acquisition of Flapping

Richter (2018) has put forward a model of the acquisition of flapping based on the Tolerance Principle (Yang, 2016). The Tolerance Principle holds that a grammatical rule may only be productive if given N candidates to which the rule could apply and e exceptions to that rule, the following relation holds:

$$(4) \quad e < \theta_N, \text{ where } \theta_N := \frac{N}{\ln(N)}$$

According to Richter, American children in acquisition of English initially assign underlying representations to flaps in accordance with the principle of Invariant Transparency, given previously in (1). Namely, children initially posit the rule of (5):

$$(5) \quad H_0: \text{ If } \textit{surface flap} \text{ then } \textit{underlying flap}$$

Eventually however, children are able to recognize some flaps as underlyingly /t/ on the basis of alternations, e.g. eating [ˈiɪŋ] as /ˈiɪtɪŋ/ by relation with the stem eat /it/. Eventually the number of exceptions to (5) of this type overwhelm the productivity threshold θ_N , and children are forced to abandon this rule in favor of an allophonic rule that captures the relationship between [t] and [ɾ]. When this happens, children begin correctly producing flaps at a lower rate, over-generalizing to [t] where [ɾ] is expected. The result is a U-shaped learning curve, as depicted in Figure 1 for Lily, a child in the CHILDES: Providence corpus.

One arguable shortcoming of the account offered in Richter (2018) is that even once children abandon the null hypothesis of (5) for an allophonic rule associating non-alternating [ɾ] with /t/ or /d/, it is unclear how

they are meant to determine which of the two coronal stops, /t/ or /d/, they should assign to a token of the surface phone [r]. Indeed, further language acquisition evidence suggests that children do not select /t/ or /d/ as URs for non-alternating flaps reliably (with respect to their orthographic values), even after they have arrived at an allophonic rule relating coronal stops to flaps. Richter (2018) herself notes tokens like *soda* and *Florida* with unexpected /t/.

Similarly, Klein & Altman (2002) observe high rates of both [t] and [d] in children aged 4;0 to 5;0. The table in (6), extracted from their paper, shows the mean rate of realizations of orthographic /t/ and /d/ as [r], [t], and [d] by the children in their study³. This indicates that even when producing stops in place of flaps, children do not reliably produce the stop that corresponds to the target UR. Indeed, for supposed underlying /t/, children are more likely to produce a [d] than a [t] (34.41% vs. 22.98%).

(6)	Spelling	[r]	[t]	[d]
	/t/	52.89%	22.98%	34.41%
	/d/	48.70%	16.36%	50.10%

Unfortunately, Klein & Altman (2002) do not indicate whether error rates of this type are different between alternating morpheme-final and non-alternating morpheme-internal conditions. However, these data do potentially reveal an interesting flaw in the blind assumption that adults represent non-alternating flaps reliably with respect to their orthographic value. There are no studies into the acquisition of flapping indicating that this confusion of stops resolves itself before children begin producing flaps categorically. In other words, we do not have any direct evidence that children (and the adult they grow into) ever do in fact learn to represent flaps faithfully to their orthographic/historic values⁴.

Indeed, some studies of incomplete neutralization of flaps report findings that are consistent with the possibility that adults continue to represent non-alternating flaps unfaithfully vis-à-vis orthographic value. In a study of New York English, Huff (1980) found that vowels were longer before /d/-flaps than /t/-flaps word finally for all vowel classes, but only for /aɪ/ word-internally. These environments are exactly those that are likely to have disambiguating evidence for the underlying value of a flap (discussed in section 2.2)⁵. In a similar vein, Herd et al. (2010) found a greater difference in pre-flaps vowel durations before morpheme-final flaps than morpheme-internal flaps (8 ms vs. 3 ms), though this finding was based on a small sample (3 pairs with morpheme-internal flaps and 5 with morpheme final-flaps) and did not reach levels of significance. Additionally, exceptional behavior of Canadian raising in words like *spider* is consistent with (though again not necessarily indicative of) confusion of the URs of /d/-flaps. Such findings that report weakening of the phonetic cues for incomplete neutralization of morpheme-internal flaps are exactly what we should expect if confusion of flap URs persists into adulthood.

2.4 Evidence for the Phonological Behavior of Flaps: Pre-Flap Vowel Durations in the Philadelphia Neighborhood Corpus

In this section, I investigate the distribution of pre-flap vowel durations before ambiguous (i.e. morpheme-internal) and unambiguous (i.e. morpheme-final) flaps with orthographic/historical /t/ and /d/ in the Philadelphia Neighborhood Corpus (PNC). The data give some limited evidence that speakers represent /t/ and /d/ flaps faithfully to their orthographic/historic value even in the ambiguous morpheme-internal case, albeit in a restricted subset of cases. Given the conclusions of 2.2, namely that there are no sufficient

³Klein & Altman (2002) note the following about the data in this chart: “the percentages in each cell do not total 100 because (a) mean % is the average of the number of times a variant was used divided by the total number of opportunities for the production of a given word-ending category, for each child, for each session, within a given age group, (b) the number of opportunities and the number of sessions were not the same for each child”

⁴One might object to this claim, given that adults generally know whether to spell a word with a *t* or a *d*. However, the supposition that such orthographic knowledge indicates that adults represent flaps faithfully to their historical/orthographic value hinges on non-trivial assumptions about the closeness of orthographic and phonological knowledge. The potential use of orthography in assigning URs will be discussed later

⁵Actually, this technically excludes on unambiguous environment, i.e. word-internal morpheme-final flaps, though Huff did not report findings in connection with morpheme-finality.

linguistic cues for flap representation in such cases, this would seem to give evidence that orthographic knowledge plays a role in representation of flaps (or at least a role in the production of words containing flaps). However, I will also explore a few reasons for taking this finding with a grain of salt.

2.4.1 Vowel Durations as a Diagnostic of Underlying Phonological Behavior of Flaps

In Section 2.2.3, we discussed how pre-flap vowel durations are not a sufficient cue for assigning /t/ or /d/ to an instance of [ɾ] since the voicing-conditioned difference between vowel durations is imperceptibly small. Even using precise measurements from speech analysis software such as Praat, which are able to detect differences of a few milliseconds, it is not possible given an token of a word with a flap to reliably classify that flap as underlying /t/ or /d/, since the distributions of vowel durations before /t/- and /d/-flaps are heavily overlapping. To do so would require a large number of tokens for each word before we could show that the mean pre-flap duration is significantly closer to the overall duration means before /t/- or /d/-flaps, and apart from certain very frequent words, we generally do not have enough data to accomplish this.

However, this does not completely void the usefulness of pre-flap vowel durations in assessing the underlying phonological behavior of flaps. In particular, we can look at pre-flap vowel durations for large groups of flaps, e.g. morpheme-internal vs. morpheme-final /t/- and /d/-flaps to determine if the faithful acquisition of /t/- and /d/-flaps is made as reliably morpheme-internally as it is morpheme-finally.

To illustrate such an approach with a similar example, consider height distribution differences between men and women. Although men are significantly taller than women, they have overlapping height distributions, and it is not possible to reliably guess the sex of an individual based solely on height. However, if we have two groups of individuals, group A composed of 1000 men and group B composed of 1000 women, an outside individual blind to the sex of the members in each group would be able to label group A as being made up of men and group B as being made up of women if he or she were given the mean height of the individuals in each group⁶. Now, suppose that there also exists a group C composed of 700 men and 300 women and a group D composed of 300 men and 700 women. An outsider would again be able to determine that group C has more men than group D on the basis of mean height. More than this, the outsider will be able to determine that groups C and D are less consistently male and female than groups A and B respectively, since the mean heights of C and D will be significantly less extreme.

In analyzing the underlying phonological behavior of groups of flaps, we can apply this same approach, replacing height with pre-flap vowel duration, men with /d/-flaps, and women with /t/-flaps. We expect morpheme-final flaps to have URs consistent with their orthographic value (so morpheme final orthographic /d/ and /t/ are analogous to the homogeneous groups A and B in the above example). What we expect for vowel durations before morpheme-internal flaps depends on whether spelling plays a role. If spelling plays no role and children in acquisition depend solely on non-metalinguistic cues (which we have shown to be an inadequate cue), then we expect no significant difference in vowel durations before morpheme-internal orthographic /t/- and /d/-flaps. If spelling plays a partial role then we also should likely expect a significant difference in vowel durations before orthographic /t/ and /d/, but the mean vowel durations should be less extreme (analogous to group C and D in the human heights example). If spelling allows for perfect or near-perfect acquisition of flaps, then we expect the distributions of vowel durations before morpheme-medial orthographic /t/ and /d/ to match these distributions before their morpheme-final counterparts.

2.4.2 Data and Methodology

The phonetic data in the current analysis were taken from the Philadelphia Neighborhood Corpus (Labov & Rosenfelder 2011), a large collection of sociolinguistic interviews of native Philadelphians conducted between the years 1973 and 2012 for the course Ling 560 at the University of Pennsylvania. All word frequency data come from the SUB-TLEX corpus (Brysbaert & New 2009).

The interview recordings in the PNC have previously been transcribed in the transcription software ELAN and force-aligned using the FAVE suite (Rosenfelder et al. 2014) to produce time-aligned Praat TextGrids,

⁶Assume for simplicity that we are controlling for other factors that influence height

which include word and phone tiers for each speaker. The intervals of the phone tiers corresponded to phonemes in the ARPAbet transcription of words taken from the CMU pronouncing dictionary. I wrote scripts in Praat in order to extract phonetic data associated with flaps, namely flap duration, preceding vowel duration, and speech rate in terms of vowels per second in the surround 7 word window, as well as information about the orthographic value of the flap, the word in which the flap occurred, and the phonemic value of the surrounding segments.

For the purposes of this study, a flap was defined as any internal segment transcribed as phonemic /t/ or /d/ that was preceded by a stressed vowel and followed by an unstressed vowel. As discussed in Section 2.1, the actual distribution of flaps word-internally is somewhat broader, but this more restricted environment was chosen i) for sake of control, ii) because some other flapping environments are sometimes considered variable for some speakers (e.g. between unstressed syllables), and iii) to follow the precedent of past corpus studies of flaps, e.g. Patterson & Connine (2001). Additionally, potential flaps were excluded before potential syllabic /n/, since this environment is known to cause /t/-glottalization, which bleeds flapping (Patterson & Connine 2001). Potential syllabic /n/ was any reduced vowel followed by a /n/ or (/ŋ/ in the dictionary transcription of a word (/ŋ/ was excluded as well because it can variably be realized as /n/ as a result of g-dropping).

Once all flaps had been extracted, a list of words containing flaps was compiled and coded for whether or not the flap occurred morpheme-finally. Morpheme-internal flaps that are involved in stress alternations (e.g. *atom* vs. *atomic*) were also coded and excluded for simplicity. The vowel phonemes /ʊ, ɔɪ, ɔ/, each of which did not occur in at least one combination of the four conditions based on orthography and morpheme-finality, were excluded.

After data extraction and elimination of unwanted tokens as described above, we were left with 2702 word types and 11201 flap tokens, including 1296 morpheme-final /t/-flaps, 7239 morpheme-internal /t/-flaps, 560 morpheme-final /d/-flaps, and 2106 morpheme-final /d/-flaps. These data were then further processed and analyzed in the statistical computing software R.

2.4.3 Results: Flap Durations in the PNC

As discussed in Section 2.2.3, past studies have generally found no significant difference between the duration of /t/-flaps and /d/-flaps themselves. The data from the PNC, however, actually do demonstrate a statistically significant difference of 9.7 ms (43.7 ms for /t/-flaps and 36.7 ms for /d/-flaps), and this difference was actually slightly greater in the morpheme-medial condition ($43.6 - 39.1 = 4.5$ ms morpheme-finally vs. $43.8 - 35.7 = -8.05$ morpheme internally). However, the interaction of orthography and morpheme-finality was not significant. When controlling for preceding and following vowel, log(speech rate), and log(frequency), the effect of orthography dropped slightly to 6 ms, but nevertheless remained significant with a t-value of 5.06 ($p < 0.001$). Although small, a 6 ms difference is still much greater than the miniscule 0.9 ms difference found by Herd et al. (2010).

Why a difference emerged here where none has emerged in past studies is not totally clear. It seems quite likely that this has to do with the much larger sample size of the current study. However, there may be some reasons for skepticism. Since the data were not checked for tokens with a non-flapped stop pronunciation, it may be possible that such tokens are the reason for this difference, skewing the distributions. It may also be the case that the FAVE-aligner is biased towards giving instances of the /t/-phoneme a longer duration based on training over samples with unflapped pronunciations, though again, this is speculation and warrants further investigation.

2.4.4 Results: Pre-Flap Vowel Durations in the PNC

While looking at vowel duration, the data were split into morpheme-final and morpheme-medial subsets and analyzed using separate linear regression models with orthographic representation as a predictor and controls for vowel phoneme, log(speech rate), log(word frequency), and following syllable nucleus. Additionally, random effects were included for both speaker and word.

Vowel	Before Morpheme-Final Flaps					Before Morpheme-Medial Flaps				
	# /t/-flaps	# /d/-flaps	Effect (ms)	SE (ms)	T-Value	# /t/-flaps	# /d/-flaps	Effect (ms)	SE (ms)	T-Value
All Vowels	1296	560	-15.591	4.931	-3.162	7239	2106	-7.619	2.748	-2.772
/ɪ/	34	10	-5.779	13.279	-0.435	3675	339	-6.882	5.853	-1.176
/ɛ/	695	31	-8.408	16.159	-0.520	329	369	2.036	5.103	0.399
/eɪ/	21	58	5.513	18.620	0.296	786	535	26.306	20.574	1.279
/æ/	12	61	-5.158	20.636	-0.250	735	121	0.688	10.693	0.064
/ʊ/	135	16	11.762	19.743	0.596	325	50	-5.100	20.474	-0.249
/i/	104	139	-12.215	10.177	-1.200	70	142	3.974	10.979	0.362
/aɪ/	177	142	-47.988	10.971	-4.374	44	88	-104.547	20.263	-5.160
/ɑ/	14	2	20.373	50.928	0.400	227	184	-12.036	6.521	-1.846
/oʊ/	24	9	19.347	12.508	1.547	230	62	-41.233	7.433	-5.547
/ʌ/	5	48	25.493	29.936	0.852	51	140	24.136	25.437	0.949
/ɜ/	67	7	25.493	29.936	0.852	64	31	24.136	25.437	0.949

Figure 2: Summarization of results of linear regression model for vowel duration on morpheme-final and morpheme-medial subsets of the data for different vowel phonemes. Effect refers to the effect of orthographic /t/ compared with /d/

The average vowel durations were as follows: 114 ms before morpheme-final /t/-flaps, 131 ms before morpheme-final /d/-flaps, 78.2 ms before morpheme-internal /t/-flaps, and 104 ms before morpheme-internal /d/-flaps. Both subsets were fitted with a linear regression model. Although the difference between /t/ and /d/-flaps was greater in the morpheme internal condition (25.7 ms vs. 17.2), after including controls in the linear regression model, the relative magnitudes of the effects reversed, with a 15.6 ms difference morpheme finally vs 7.6 ms morpheme internally.

At this point we might draw several conclusions. Firstly, the fact that the difference between flap durations is significant even in the morpheme-internal condition seems to indicate that spelling must play some role in the representation of flaps, or at the very least in the production of flaps. Secondly, the fact that the vowel duration difference is smaller under the morpheme-internal condition is consistent with the proposition that some amount of confusion between /t/- and /d/-flaps persists into adulthood, and that orthography, although relevant, is not a perfect means by which to learn the underlying representation of a flap.

However, there are some problems with these conclusions. When we combine the data into a single linear regression model that also includes controls for morpheme-finality and a term for the interaction of morpheme-finality and orthographic value, this interaction term does not reach levels of significance ($p = 0.16$). Granted, it is still possible to have a relatively high degree of confusion without this interaction term becoming significant.⁷ However, we are still unable to reject the hypothesis that there is no underlying difference between historic /t/ and /d/ in the morpheme-internal and -final conditions. Hence, this finding is consistent with but not indicative of some degree of confusion of URs.

However, the picture becomes more complicated when we split the two conditions (morpheme-medial and morpheme-final) by vowel phoneme. For each vowel phoneme, I again fitted a separate linear regression model under both conditions with orthographic representation as a predictor and controls for log(speech rate), log(word frequency), and following syllable nucleus (this includes all the previous controls aside from vowel phoneme, which is obviously uniform within each vowel phoneme category). The results of these models can be seen in Figure 2, with statistically significant observations printed in bold face (all have $p < 0.01$ except for /ɑ/ in the morpheme-medial condition, which has $p < 0.05$).

When split up by vowel phoneme in this way, we see that the statistical significance of duration differences by-and-large collapses, in spite of large sets of data in many cases. Indeed, the effect of underlying voicing on vowel duration goes in the wrong direction for a majority of vowels, though none of the effects in the

⁷To test this, I ran a simulation of vowel durations before morpheme-internal flaps using different confusion rates. Drawing pre-flap vowel durations from idealized normal distributions with the same means and standard deviations as the observed distributions in the morpheme-final flap condition, I found that the interaction term did not reach even a loose level of significance $p = 0.10$ until 50% of tokens in both the orthographic /t/ and /d/ conditions were randomly assigned underlying /t/ or /d/ (both with 50% probability). This would seem to indicate that up to a quarter of flaps may be incorrectly assigned in the morpheme-internal case before the interaction term becomes significant.

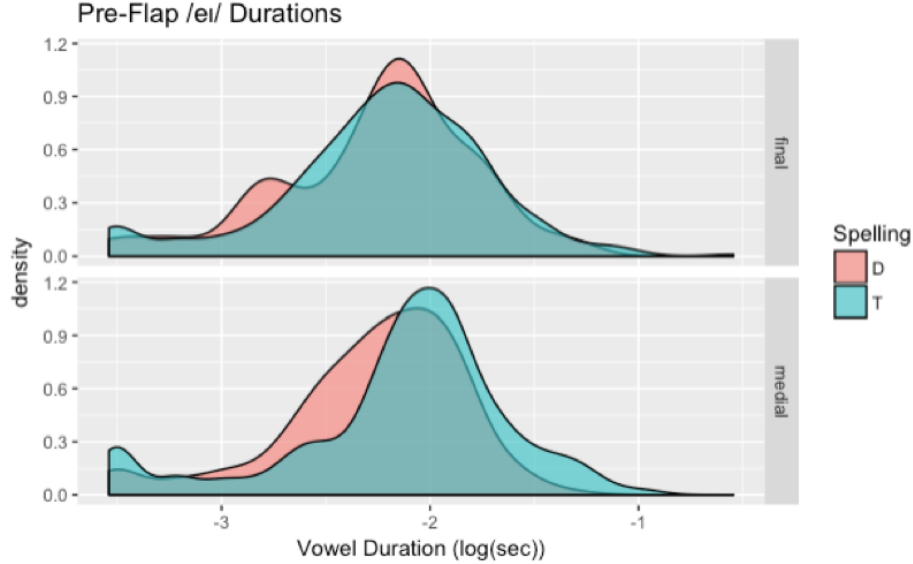


Figure 3: Pre-flap /eɪ/ durations

unexpected direction are significant. Figure 3 shows a typical vowel distribution with no significant difference in either condition. Indeed, only /aɪ/ shows a significant duration difference in both conditions, and /ɑ/ and /oʊ/ both show differences in the morpheme-medial, but not the morpheme-final conditions.

The behavior of /aɪ/ is not altogether surprising, since Philadelphians participate in Canadian raising and therefore have a distinct raised allophone [aɪ] before underlyingly voiceless consonants, which might be expected to have a shorter duration. Furthermore, the quality difference of the nucleus of the /aɪ/ diphthong might also provide a sufficiently salient cue for identification of the underlying voicing of a flap not shared by other phonemes (as discussed in Section 2.2.3), potentially accounting for this phonemes distinct behavior. This behavior of /aɪ/ is also parallel to Huff (1980)’s finding of significant pre-flap duration differences only before /aɪ/ word-internally in New York speech. Figure 4 shows the distributions of pre-flap /aɪ/ tokens under the different conditions.

When we exclude /aɪ/ from the data used in the linear regression models that include all vowel morphemes together, the difference between vowel durations before /t/- and /d/-flaps falls below levels of significance in the morpheme-final condition, but not in the morpheme-internal condition. This is the exact opposite of what we would expect. However, we note that ratio of morpheme-final to morpheme-internal flaps is about 1:5 (1296 /t/-flaps and 560 /d/-flaps for a total of 1856 flaps in the morpheme-final condition vs. 7239 /t/-flaps and 2106 /d/-flaps for a total of 9336 under the morpheme-internal condition). Thus, it seems likely that we can attribute this unexpected behavior to the difference in *n*-values.

Why /ɑ/ and /oʊ/ should also be unique in having statistically significant duration differences in the morpheme-medial condition is more difficult to account for than the behavior of /aɪ/ without referring to the influence of spelling on UR assignment. Again, it may also seem unexpected that these two vowel classes show differences in the morpheme-medial, but not the morpheme-final condition (which is expected to have more evidence for UR), though once again, this can likely be attributed to the low sample size in that condition (/ɑ/ only had 14 /t/-flaps and 2 /d/-flaps under the morpheme-final condition, while /oʊ/ only had 24 /t/-flaps and 9 /d/-flaps).

Even if the lack of a statistically significant difference between pre-flap vowel durations for the phonemes /ɑ/ and /oʊ/ under the morpheme-final condition can be explained on the basis low sample sizes, the fact that they do show statistically significant differences in the morpheme-medial condition is somewhat puzzling, independent of the behavior of these vowels in the morpheme-final condition. To reiterate, it does not seem that there is any reliable cue for the underlying voicing of flaps in this condition aside from spelling.

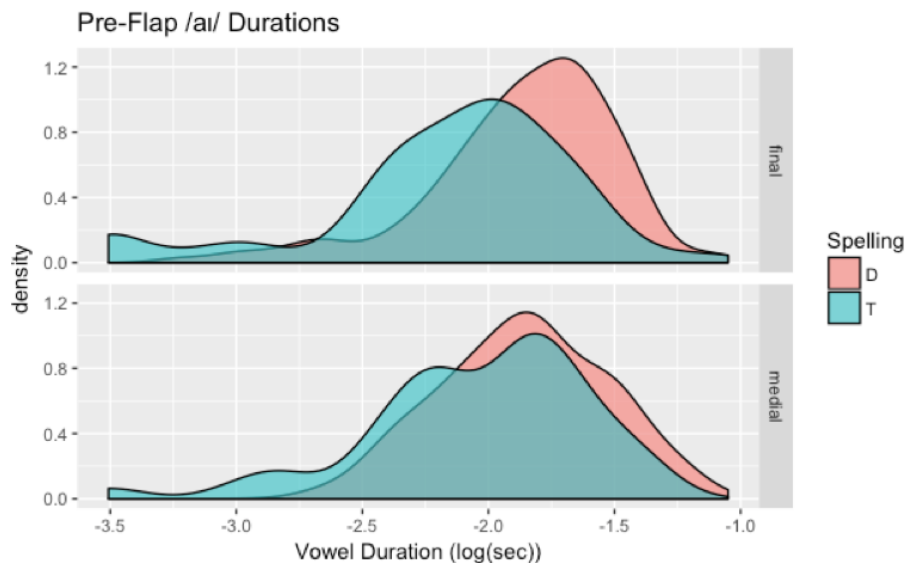


Figure 4: Pre-flap /aɪ/ durations

However, if spelling does allow for the faithful acquisition of morpheme-internal flaps after these two vowels, why do we not observe similar behavior in morpheme-internal flaps after other vowel phonemes, particular those with a larger sample size? A search for multicollinearities did not find any clear reason to suspect that the differential behavior of durations for these two vowels is a quirk of the statistical model. Further investigation, perhaps in a controlled experimental setting, is likely warranted. The distribution of pre-flap /oʊ/ durations are displayed in Figure 5.

2.4.5 Results: Pre-Flap Vowel Durations in Minimal Pairs within the PNC

The only study discussed previously in this work that investigated pre-flap vowel durations within a corpus (rather than through experimental approaches) was Patterson & Connine (2001), which analyzed flaps in the SWITCHBOARD Corpus. Rather than look at vowel durations before all of the flaps in the entire corpus, the authors mined the corpus for word pairs that differed from each other only in whether they had a medial /t/ or /d/. The search found 11 such pairs, with 60 tokens of medial /t/-words and 55 tokens of medial /d/-words. The average difference was found to be 16 ms (111 vs 95 ms), with a significance of $p < 0.05$. In this section, I have replicated this approach within the PNC.

In total, 20 such pairs were found and there were 59 /d/ tokens and 334 /t/ tokens. The difference between the mean vowel durations was a much larger 53 ms (155 vs. 102 ms). In a linear regression model, this effect reduced to 43 ms when controlling for $\log(\text{frequency})$ and $\log(\text{speech rate})$ as controls, which was nevertheless still significant with a t-value of -4.676 ($p < 0.001$). The fact that the difference found here is so much greater than when looking at the PNC as a whole is likely a consequence of the fact that these data are more controlled (since they are homophones) and that a significant portion of the variation in the larger data sample resulted from some set of predictors that were not identified and captured by the statistical models used.

The results for each individual word pair are displayed in Figure 6. We note that some pairs have differences in the opposite of the expected direction, though the trend is overwhelmingly for vowels to be longer before /d/-flaps than /t/-flaps. Figure 7 visualizes this difference with a density plot for all tokens that were included in any of the homophone pairs. This figure clearly shows the two distinct distributions for vowels before orthographic /t/ and /d/ in both morpheme-finality conditions. Additionally, there does not appear to be much of a difference between these two distributions under the two conditions; running a linear regression model with an interaction term for morpheme-finality and orthographic value gives a p -value well outside

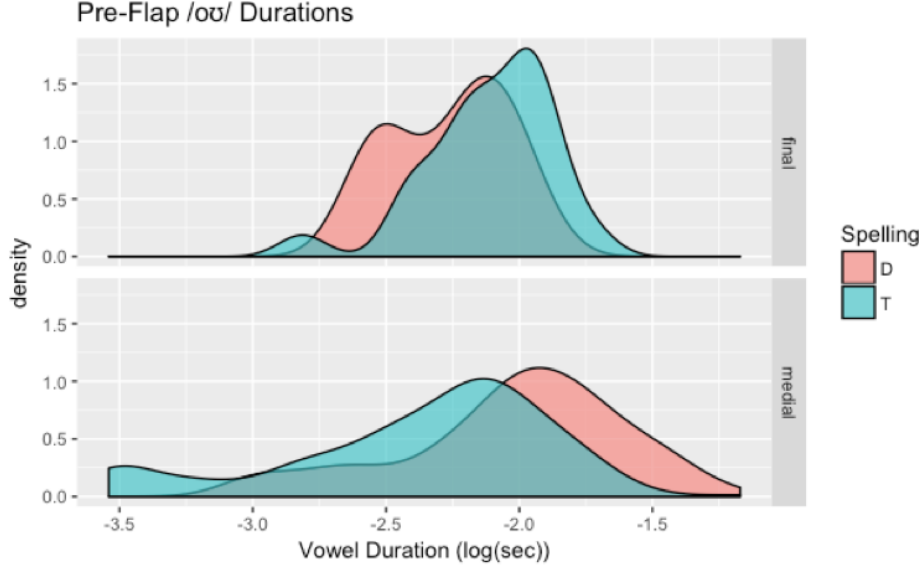


Figure 5: Pre-flap /ou/ durations

common significance thresholds ($p = 0.66$). This seems to give stronger evidence that individuals represent flaps faithfully to their orthographic value, which in turn would suggest that spelling indeed plays a role (given the insufficiency of phonetic cues).

2.4.6 Pre-Flap Vowel Durations as a Voicing Cue Revisited

Based on the relative crispness of the voicing-conditioned duration differences in the homophone data of Section 2.4.5 compared with corresponding data for all flap words in Section 2.4.4, we might be more inclined to posit that children are able to make use of pre-flap vowel duration as a cue for flap URs. However, children do not face the task of identifying URs only of those ambiguous flaps that happen to be parts of homophonous pairs, but also of those in words not belonging to homophonous pairs, so it still poses a problem to the position that preceding vowel durations is a usable cue for the child in acquisition if it only works well for a neat, restricted subset of the linguistic stimulus. If sophisticated statistical models that are able to keep track of duration differences of mere milliseconds for large numbers of tokens are unable to find significant differences between durations before /t/ and /d/-flaps, can we really reasonably expect a child to so?

Even supposing that children are able to keep track of exact vowel durations in milliseconds for all observed tokens of any given word (which seems dubious), again, it still appears that they would be unable to faithfully acquire flaps on the basis of this duration cues alone. Consider a simple algorithm for assigning an underlying representation to an ambiguous flap in a given morpheme based on whether the average pre-flap vowel duration for all tokens of that morpheme falls closer to the mean of unambiguous pre-/t/-flap or pre-/d/-flap vowel durations. This method overwhelmingly prefers /t/ for both orthographic *t* and *d*, likely owing to the fact that vowels tend to be shorter (and hence more pre-/t/-like) before morpheme-internal flaps. Applying this approach to the PNC, the overall ‘success’ rate is 80.3% for orthographic /t/ and 31.2% for orthographic /d/, for an overall success rate of 69.7%. Even if we can improve this approach by increasing the sample size and controlling for other factors that influence vowel duration, it remains clear that it will not be possible to achieve a 100% success rate.

If we suppose that children in acquisition do indeed use pre-flap vowel durations to assign URs to ambiguous morpheme-medial flaps, albeit imperfectly (and furthermore that spelling plays no role), then the fact that the learning of URs this way will involve some percentage of errors has the consequence that the errors will add noise to the phonetic signal that children of subsequent generations use to make UR assignments. Hence, errors should propagate over generations, causing the phonetic cues of a faithful differential acquisition of

/t/-flaps		/d/-flaps		Difference (ms)
Word	Vowel Duration (ms)	Word	Vowel Duration (ms)	
BATTER	225.5	BADDER	200.0	-25.5
BITTER	96.7	BIDDER	70.0	-26.7
BRATED	90.0	BRAIDED	260.0	170.0
CATTY	120.0	CADDY	180.0	60.0
DUTY	121.0	DOODY	216.0	95.0
GREATER	80.5	GRADER	115.6	35.1
HATTER	174.0	HADDER	180.0	6.0
KITTY	50.0	KIDDIE	40.0	-10.0
KITTY	50.0	KIDDY	89.0	39.0
LATTER	116.7	LADDER	177.8	61.2
MATTER	101.0	MADDER	181.4	80.4
METAL	79.1	MEDAL	130.0	50.9
METAL	79.1	MEDDLE	100.0	20.9
METALS	60.0	MEDALS	80.5	20.5
METALS	60.0	MEDDLES	110.0	50.0
PATTY	133.8	PADDY	200.0	66.3
SEATER	90.0	CEDAR	90.5	0.5
WRITER	104.5	RIDER	154.2	49.7
WRITER	104.5	RYDER	160.0	55.5
WRITERS	135.5	RIDERS	155.0	19.5

Figure 6: Pre-Flap Vowel Durations in homophone pairs. Positive differences are in the expected direction

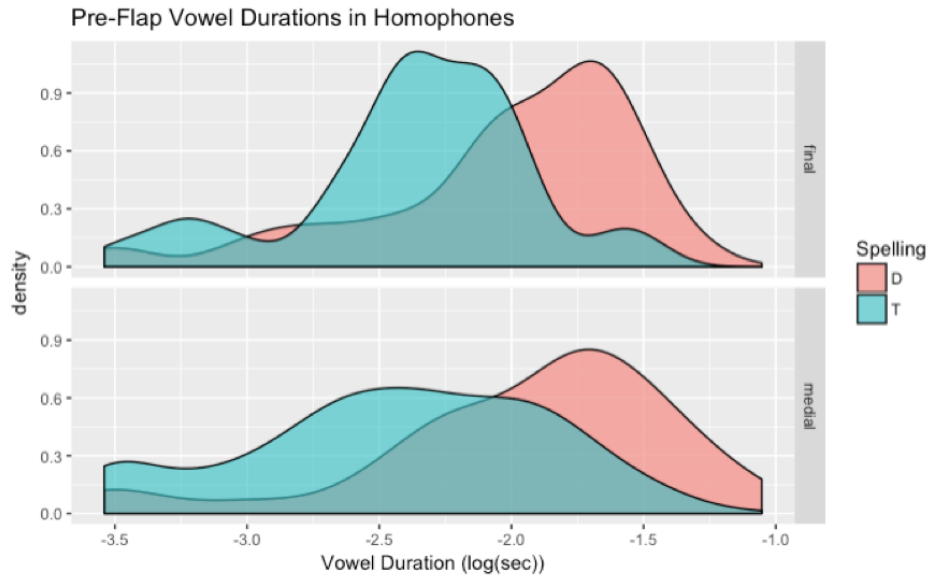


Figure 7: Pre-flap vowel duration distributions in homophones

/t/- and /d/-flaps to erode over time.⁸ The fact that we still observe evidence that flaps are represented faithfully to their orthographic value even many generations after the inception of intervocalic flapping in American English, since which we should expect to observe extensive error propagation of the sort described above resulting in an eroded opposition between ambiguous /t/- and /d/-flaps in current speech, gives more evidence that pre-flap vowel durations alone cannot be the means by which individuals acquire flap URs.

2.5 Section Summary and Discussion

The facts presented in the preceding subsections of Section 2 give a complicated picture of the phonological behavior of ambiguous phones resulting from conditioned mergers in cases where neither an obvious default rule for UR assignment nor sufficient disambiguating evidence for the underlying representations used by the preceding generation exists.

Firstly, we can see that such scenarios do indeed occur, as exemplified by the case of intervocalic flapping in North American English. In particular, we have shown that (most) morpheme-internal flaps in the / $\check{V}$ _V/ environment fall into this ambiguous category, as 1) they do not alternate with unambiguous stops due to environmental changes, 2) they do not alternate with unambiguous stops due to variable rule application since flapping is categorical in this environment, and 3) the incomplete neutralization of flaps provide inadequate phonetic cues of the underlying voicing of an underlying flap, rendering reliable transmission of historically/orthographically-faithful URs over generations by this means implausible.

Secondly, although it is typically supposed that flaps are represented faithfully to their orthographic/historic value by adults, previous research into the acquisition of flapping does provide any robust evidence in support of this assumption. If anything, we have evidence that children are actually quite bad at picking out the ‘correct’ representations of flaps, showing a high degree of use of the non-etymological coronal stop (i.e. /t/ for historic /d/ and vice versa) during the period that they still do not use flapping categorically. The fact that non-flapped realizations do not appear to come in line with their orthographic values before the start of categorical flapping suggests that we should adopt a null hypothesis that this confusion of flap URs vis-à-vis their historic/orthographic value persists beyond acquisition into adulthood.⁹ Some past research into incomplete neutralization of flaps is consistent with (though not necessarily indicative of) this hypothesis, showing weaker differences between supposed /t/- and /d/-flaps morpheme-internally. Observed facts about the interaction of flapping with allophonic rules of adjacent segments counterbled by flapping, namely a high rate of ‘exceptions’ to such rules in the case of Canadian raising, are also consistent with this assumption.¹⁰

Thirdly, we saw how differences in pre-flap vowel durations associated with incomplete neutralization of flaps, although unreliable as a means of identifying how individual flap tokens are represented by their speaker, allow us to investigate the general phonological behavior of flaps from historic /t/ and /d/ under ambiguous and unambiguous conditions.

Applying this approach to pre-flap vowel durations in the Philadelphia Neighborhood Corpus gave a mixed bag of results. An analysis of all vowels in the PNC (excluding vowels phonemes with no tokens in one of the four conditions), found a statistical difference between vowel durations before historic /t/ and /d/ in both the ambiguous and unambiguous condition, but the significance of these differences by-and-large collapsed when we looked at individual phonemes. Only differences in /aɪ/ durations were statistically significant in both the unambiguous and ambiguous conditions, likely owing to the allophony exhibited by this phoneme

⁸Perhaps in a similar fashion to erosion as modeled in Section 3.1

⁹Anecdotal evidence seems to support this conclusion. I have peers who report hearing others use realizations like [ˈæti] for *ready* when hyper-articulating or attempt to imitate an accent that lacks flapping

¹⁰The fact that at least some speakers, myself included, allow Canadian raising before all morpheme-internal flaps combined with the fact that vowel durations before supposed /d/-flaps fall into the range of pre-voiceless vowel durations might lead us to explore the idea that ambiguous flaps are simply assigned /t/, where generalization in this direction is perhaps motivated by the fact that unambiguous morpheme-final /t/-flaps are more frequent than /d/-flaps. However, we have only limited evidence to investigate this possibility. Fruehwald (2016) finds that ‘exceptional’ Canadian raising before orthographic /d/-flaps developed in Philadelphia only in second half of the 20th century, but if all ambiguous flaps are underlyingly /t/, then we would expect Canadian raising before flaps from historic /d/ to develop concurrently with Canadian raising in other segments during the first half of the 20th century. This piece of counter-evidence, as well as the fact that unflapped /d/ is a more common realization of expected flaps among children in acquisition for historic /t/ as well as /d/ draws the plausibility of a generalized [ɹ] = /t/ rule into question

through Canadian raising. Removing /aɪ/ from our statistical models, overall duration differences were no longer significant before morpheme-final flaps, even as they remained significant before morpheme-medial flaps. This unexpected result likely resulted from differences in sample sizes. Additionally, the differences in the morpheme-medial group appear to be based on a limited subset of the data; the only individual vowel phonemes to show significant differences in duration before historic /t/ and /d/ were /ɑ/ and /ou/, somewhat mysteriously. These data weakly suggest that faithful representation of flaps from historic /t/ and /d/ occurs even in the ambiguous morpheme-medial condition, suggesting that spelling must play a role. Additionally, the difficulty of finding significant duration differences before flaps in the relatively naturalistic speech data of the PNC, even when we make use of large sample sizes, precise measurements, and relatively sophisticated statistical models reinforces our claim that vowel duration differences cannot plausibly be used by children to discern underlying representations.

Interestingly, when we limit the tokens used in our sample to words found in homophonous pairs, the statistical differences become much crisper, with statistically significant differences holding for both the ambiguous morpheme-internal condition and the unambiguous morpheme-final condition even with much lower sample sizes. This more controlled analysis gives much stronger evidence that individuals do indeed represent flaps faithfully to their orthographic value even in the absence of strictly non-metalinguistic disambiguating evidence, suggesting that spelling plays a role in the production if not the representation of flaps.

This leaves us with an interesting conversation to be had regarding the relationship between orthographic and phonological knowledge. On one hand, given well-documented phenomena like spelling pronunciations, a close relationship may not seem wholly unreasonable. On the other hand, such an assumption seems unlikely given the ways that orthography and phonology frequently diverge. Furthermore, the idea that spelling is a powerful enough cue of underlying representation that it can allow for reliable, faithful transmission of ambiguous URs across the entire speech community over many generations would seem to potentially contradict several generally accepted principles of language change, e.g. that mergers are irreversible under the normal mechanism of sound change (if two phonemes are merged but have distinct spellings, as is often the case, then speakers can simply consult their orthographic knowledge to reverse the merger). There is much more to be said and debated this discussion, but this is beyond the scope of the current work.

Moreover, we are left with a question of how the phonological behavior of flaps (or other conditioned phones with similar properties) would pattern without the influence of orthography. If spelling indeed plays a major role in representation and/or production of phonemically ambiguous merged phones, then what patterns of phonological behavior could we expect for example among illiterate or for languages whose orthographic systems do not reflect the historic contrast between the phonemes implicated in the merger? We leave such questions open to future work.

3 Limitations of Disambiguating Evidence: Erosion of Liaison Systems

Section 2 explored the phonological behavior of merged phones in situations marked by a lack of disambiguating evidence for how those phones should be underlyingly represented. However, phonological systems resulting from conditioned mergers show at times unusual patterns of diachronic development and synchronic behavior even in the presence of potential disambiguating evidence.

In particular, such systems tend to demonstrate erosion of the historic opposition between the phonemes implicated in the merger. One possible explanation for why such erosion occurs may come from the intrinsic limitations of disambiguating evidence of the sort discussed in the Introduction. More precisely, erosion might result from the inherent difficulty of making decisions about the phonological representation and behavior of morphemes on the basis of limited disambiguating evidence that occurs with but a subset of the instances of that morpheme. Is it possible that some morphemes containing merged phones do not occur with the requisite disambiguating evidence early or frequently enough in the linguistic stimulus for children to reliably acquire the historically expected phonological representation and behavior of that morpheme? This raises the question of (7).

- (7) Can we prove that erosion after conditioned merger is an expected and indeed inevitable result of internal properties of the phonological system in which it occurs?

The next section (Section 3.1) attempts to answer (7) in the affirmative by putting forward a mathematical model of erosion after conditioned mergers.

In exploring erosion of phonological systems resulting from conditioned mergers, we will focus on two different liaison systems, namely English /ɹ/-liaison and French liaison. Liaison systems arise when morpheme-final consonant deletion (i.e. merger with $\emptyset$) may be bled by resyllabification with a following vowel-initial word, resulting in the apparent insertion (from a synchronic perspective) of otherwise absent ‘liaison consonants’ morpheme-finally in pre-vocalic contexts. Hence, the origin of both English /ɹ/-liaison and French liaison are quite similar. The main difference is that English /ɹ/-liaison, as the name would indicate, involves only one liaison consonant, i.e. /ɹ/, whereas French liaison involves a much broader array of consonants, in particular /z, t, n/ and to a lesser extent /p, ʁ, g/ (though historically even more consonants were involved).

Despite their similarities, the diachronic development of the two systems after their inception has differed sharply. In English, the phones /ə, ɔ:, ɑ:, ɜ:/ (which are the only phones that occur morpheme-finally in words with historic final /ɹ/ in most dialects)¹¹ may be followed by a liaison /ɹ/, regardless of whether this /ɹ/ is etymological or not (the more widely used terms linking /ɹ/ and intrusive /ɹ/ are used to refer to etymological and non-etymological /ɹ/-liaison respectively). Even though this represents an expansion of the original environment of /ɹ/-liaison, I still refer to it as erosion since it represents a loss of the historical contrast between /ɹ/-final and vowel-final morphemes. The historical development of French liaison is somewhat more complicated, though the primary trend has been loss of the system. Although French liaison purportedly could occur for any word-final consonant in any morpheme historically, it now only occurs in a very restricted set of morphemes, specifically function words, inflectional suffixes, and a small group of adjectives and adverbs, as well as in a few set frozen expressions. Section 3.4 attempts to explain the direction of erosion in these two cases using the principle of Invariant Transparency and the Tolerance Principle, already given in (1) and (4) respectively.

A further difference between the two cases is the degree of erosion. In the English case, there appears to have been total erosion of the historic contrast between /əɹ, ɔ:ɹ, ɑ:ɹ/ and /ə, ɔ:, ɑ:/, whereas French liaison has undergone only partial erosion, and the historic contrast still exists, albeit in a restricted subset of environments vis-à-vis its historical distribution. We may be able to explain this difference in degree of erosion if it can be shown that it is easier for children to learn French liaison for the morphemes where it still occurs than to learn exceptions to linking-/ɹ/. Furthermore, we may be able to explain why French liaison has been robust against erosion in some morphemes but not in others if we can show that liaison is easier to learn for morphemes or groupings of morphemes where it currently does occur than for morphemes where it currently does not. Both of these problems will be explored in Section 3.5 on the basis of the mathematical erosion model of Section 3.1 as well as facts about the two liaison systems in question.

3.1 Mathematical Modeling of Erosion after Conditioned Merger

3.1.1 The Model

Consider some conditioned merger of two phonemes a and b to the surface value c in the set of environments M (c may or may not be phonetically identical to unmerged surface a or b)¹². Then, the underlying value of surface c in M is ambiguous between a and b . Let us assume some general rule R by which the child assigns underlying a rather than b to c as a default in the absence of sufficient disambiguating evidence.¹³

¹¹Note that there are a number of dialects historically with centralizing diphthongs ending in /ə/ which may have monophthongized realizations and which thus would no longer be included in this set, e.g. /ɛ:/ for /ɛə/. These are still included in the set of vowels that may be followed by a liaison /ɹ/-consonant

¹²Note that it is possible for one of $\{a, b\}$ to have the underlying value $\emptyset$ in cases of deletion or insertion (the case in both of the erosion cases explored in the current work). Although $\emptyset$ is technically not a phoneme, the word phoneme will be used so as to include $\emptyset$ for sake of simplicity.

¹³Section 3.4 identifies a general rule R for both of the liaison systems considered in the current work. In cases where we cannot identify any default rule R , then intuitively we might still expect erosion to occur, but the modeling of such erosion

Now consider some morpheme m containing an instance x of some phoneme in $\{a, b\}$, such that x alternates environments between the sets of environments M' and $D = M'^C$, where M' is the environment where the merger occurs categorically (and x is realized ambiguously as c) and D is the environment where disambiguating non-merged alternants may occur (i.e. the union of variably merging environments and categorically non-merging environments). We see $M' \subseteq M$ and $M^C \subseteq D$. D may overlap with M (but never with M') due to the existence of variably merging environments.

We are interested in the probability of an exception to R being learned by the end of acquisition, i.e. the probability that b rather than a is assigned to x . Let $p^{(g)}$ denote this probability for a child in the g th generation after the merger.

On our way to estimating the value of $p^{(g)}$, we will define $p_n^{(g)}$ to denote the probability of a child in the g th generation after the merger assigning b to x after n words of linguistic input. Consider two cases:

- Case 1: The target UR of x is a . Since the grammar assigns a to x anyway by R , in the usual case we have $p_n^{(g)} = 0$ and hence $p^{(g)} = 0$ regardless of whether disambiguating evidence occurs in the stimulus¹⁴
- Case 2: The target UR of x is b . In this case, the child will only correctly assign b to x if there is a sufficient amount of disambiguating evidence in the stimulus, i.e. some threshold of $\theta \geq 1$ instances of x realized as b . Let $X_n^{(g)}$ be a random variable denoting the number of instances of x realized as b in the first n words of the linguistic input for a child in the g th generation. Then, we have $p_n^{(g)} = P(X_n^{(g)} \geq \theta)$. Let $p_{n,\theta}^{(g)}$ denote this probability for any given estimate of θ

Let us focus on the second case. Hence, we will assume that the target of x is b . It turns out that we can reasonably estimate $p_{n,\theta}^{(g)}$ and that we can use this estimate to approximate $p^{(g)}$. To do this, we define the following parameters (in addition to the parameter θ defined above):

- S : the number of words in a child's linguistic stimulus during the period of time that a child is amenable to changes in their URs. For simplicity we will assume S is constant over generations and across speakers.
- $p_e^{(g)}$: the probability that any given word contains an instance of x realized as b in the stimulus available to a child in generation g (which may be used as evidence for the UR)

We can see that $p^{(g)} = p_S^{(g)}$, which is equivalent to $p_{S,\theta}^{(g)}$ for whatever the true value of θ is. Also, we observe that $X_n^{(g)}$ follows a binomial distribution $B(n, p_e^{(g)})$ (and hence $X_S^{(g)} \sim \text{Bin}(S, p_e^{(g)})$). Thus we get the following for $p^{(g)}$:

$$\begin{aligned} p^{(g)} &= p_{S,\theta}^{(g)} \\ &= P(X_S^{(g)} \geq \theta) \\ &= 1 - P(X_S^{(g)} < \theta) \\ &= 1 - \sum_{k=0}^{\theta-1} P(X_S^{(g)} = k) \\ &= 1 - \sum_{k=0}^{\theta-1} \binom{S}{k} (p_e^{(g)})^k (1 - p_e^{(g)})^{S-k} \end{aligned}$$

becomes more difficult. For simplicity, we will not consider such cases.

¹⁴Occasionally the situation may be a bit more complicated. For example, we could conceivably have $p_n^{(g)} > 0$ if the following three conditions hold, though this is a somewhat marginal case (though it occurred historically in non-rhotic dialects of English with linking-r for instance):

1. R assigns a to x by default
2. The merged phone c is equivalent to the phonetic value of unmerged b
3. D includes variably merging environments

Since $\theta \geq 1$, $p_e^{(g)} < 1$, and S is finite, we can see that $p^{(g)} < 1$. Note, however, that the difference between 1 and $p^{(g)}$ may be vanishingly small for sufficiently large values of $p_e^{(g)}$ and N and a sufficiently small $\theta \ll Sp_e^{(g)}$, which is likely the cases for morphemes that are robust against erosion.

The inequality $p^{(g)} < 1$ also holds true when $g \leq 0$, i.e. for a child in any generation up until the one that innovated the merger. However, intuitively this results not from a non-zero chance of the child assigning a to x , but rather from the chance that the morpheme m (and hence its component phoneme x) is not learned at all; all instances of m in a linguistic input spoken by speakers who do not have the merger should have an unambiguous instance of b for x , so it would not make sense for a child in a generation whose parents lack the merger to assign a to x . For this reason and sake of simplicity, we will assume that $p^{(g)} = p_{S,\theta}^{(g)} = 1$ for all $g \leq 0$ for all possible values of θ .

As for generations $g \geq 1$, the fact that $p^{(g)} < 1$ means that there is a non-zero probability of x being erroneously assigned a , and intuitively these errors will propagate over generations and effect $p_e^{(g')}$ for all generations $g' > g$. In fact, we can identify a reasonable upper bound $p_e^{(g)} \leq p_e p^{(g-1)} = p_e p_{S,\theta}^{(g)}$ where we define $p_e = p_m p_D p_f$ for the following probabilities (which we can easily estimate from corpora):

- p_m : probability that any given word in the stimulus contains an instance of m (and hence of x). For simplicity, assume p_m is constant across generations.
- p_D : probability of x occurring in the potentially non-merged environment D given an instance of m . For simplicity, assume p_m is constant across generations.
- p_f : probability of a faithful unmerged alternant of b occurring given an instance of x in D and that the speaker of that instance of x has assigned b to x .

The reason this is an upper bound is that children can be expected to influence the speech of one another. Clearly, $p_n^{(g)} < p_S^{(g)} = p^{(g)}$ for all $n < S$. In other words, the probability that a child assigns b to x during acquisition is strictly less than this probability at the end of acquisition. We can also reasonably assume that each generation is at most as likely as the last generation to acquire x as b , i.e. $p^{(g-1)} \leq p^{(g)}$, giving us $p_n^{(g)} < p^{(g-1)}$ for all $n < S$. Suppose for simplicity that all linguistic stimulus comes from speakers who acquired the language no more than one generation prior¹⁵. Then the probability that any given speaker in the linguistic input assigns b to x — including other speakers in the process of acquisition — is capped at $p^{(g-1)}$.

The bound $p_e^{(g)} \leq p_e p_{S,\theta}^{(g-1)}$ allows us to recursively define an upper bound on $p^{(g)}$ precisely defined in terms of (θ, p_e, S, g) on the basis of the distribution $Bin(S, p_e p_{S,\theta}^{(g-1)})$:

$$p^{(g)} = p_{S,\theta}^{(g)} \leq \begin{cases} 1 & \text{if } g \leq 0 \\ 1 - \sum_{k=0}^{\theta-1} \binom{S}{k} (p_e p_{S,\theta}^{(g-1)})^k (1 - p_e p_{S,\theta}^{(g-1)})^{S-k} & \text{if } g \geq 1 \end{cases}$$

At this point, we can also note that the true value of θ likely rises over time. The reasoning is as follows: once some rising proportion of speakers in the speech community have assigned a to x , there will be a corresponding rise in the proportion of tokens unambiguously pointing to a UR of a rather than b in the set of unambiguous environments within D . In order to offset the influence of this effect on whether the speaker assigns a or b to x , the amount of evidence unambiguously pointing towards a UR of b must likely be higher as well, requiring a higher threshold θ . Thus, the upper bound on $p^{(g)}$ defined above is likely even looser than already indicated.

Depending on the parameters p_e , θ , S , and g , we can see that $1 - \sum_{k=0}^{\theta-1} \binom{S}{k} (p_e p_{S,\theta}^{(g-1)})^k (1 - p_e p_{S,\theta}^{(g-1)})^{S-k}$, and hence $p_{S,\theta}^{(g)}$, can have any value in the range $(0, 1)$. There are three cases:

¹⁵Intuitively, this is not true, but the presence of any older speakers in the input is likely offset by the presence of younger speakers. If the effect is not offset, then we can redefine the intervals of g as the age difference between the learner and the youngest older speakers in the stimulus such that the effect is offset, and this would fix the problem, though we will assume for simplicity that this is not an issue

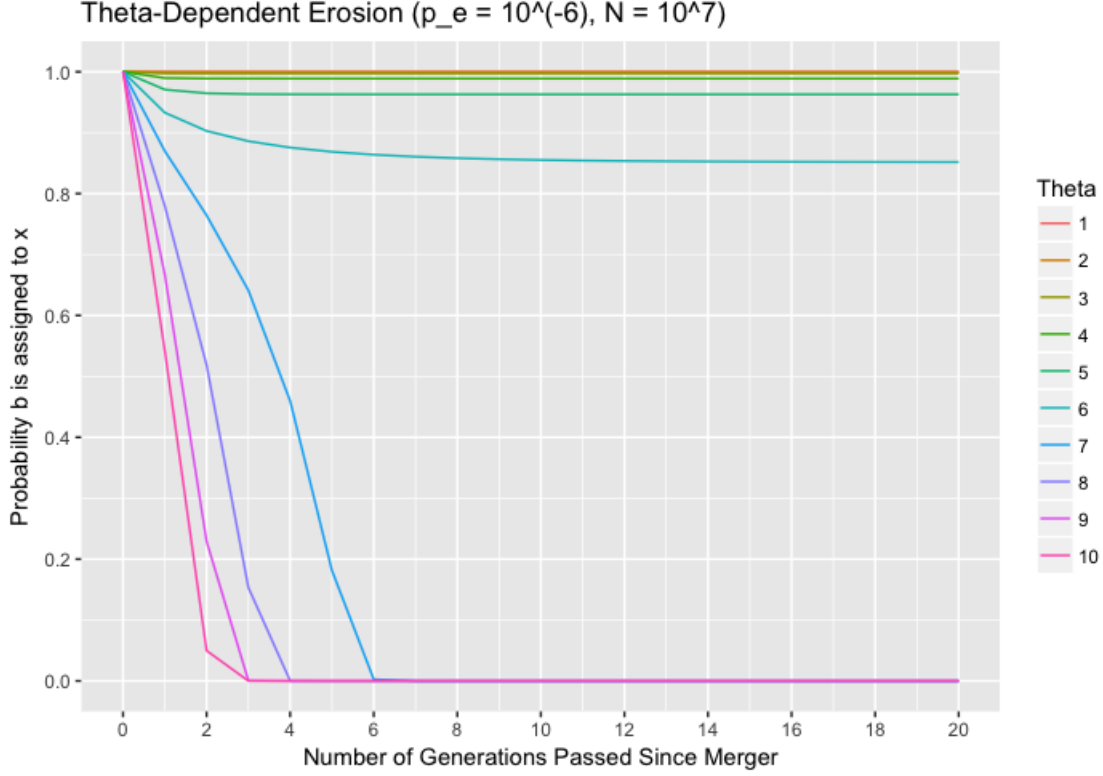


Figure 8: Simulation of upper bound of $p_{10^7, \theta}^{(g)}$ with $p_e = 10^{-6}$

- Case 1: $p_{S, \theta}^{(g)} = p_{S, \theta}^{(g-1)}$. Then, the probability of an individual assigning b to x is at a stable equilibrium across generations
- Case 2: $p_{S, \theta}^{(g)} < p_{S, \theta}^{(g-1)}$. Then each generation is less likely to assign b to x and this exception to the rule R will erode until an equilibrium as in case 1 is reached, potentially though not necessarily when $p_{S, \theta}^{(g)} = 0$ and a is assigned to x by all new learners.
- Case 3: $p_{S, \theta}^{(g)} > p_{S, \theta}^{(g-1)}$. This case does not normally occur. If the tendency of $p_{S, \theta}^{(g)}$ is to move up over generations, then $p_{S, \theta}^{(g)}$ would never have fallen below the initial value of $p_{S, \theta}^{(0)} = 1$, at which it would already have had an equilibrium¹⁶

From these cases, we see that the probability of x being assigned a UR of b erodes over time until some equilibrium is reached. Define this equilibrium value q as follows:

$$q = \lim_{g \rightarrow \infty} p^{(g)}$$

Additionally, we'll define $q_{S, \theta}$ as the value of q for any given estimate of S and θ . q is also dependent on p_e , but the real value of this parameter is easier to estimate for a given morpheme.

Here we have two cases: 1) $q = 0$ and 2) $q > 0$. A simulation of $q_{10^7, \theta}$ with $p_e = 10^{-6}$ in R involving both of these cases for different θ -values is visualized in the Figure 8. There are two things to note at this point:

- Since this simulation is based on the upper bound of $p_{S, \theta}^{(g)}$, our simulation also gives an upper bound of $q_{S, \theta}$, and the number of θ values for which $q_{S, \theta} = 0$ may actually be greater.

¹⁶Actually, this case could occur if p_e changes over time, e.g. if an infrequent word becomes much more frequent from one generation to the next. For simplicity we will ignore such marginal cases

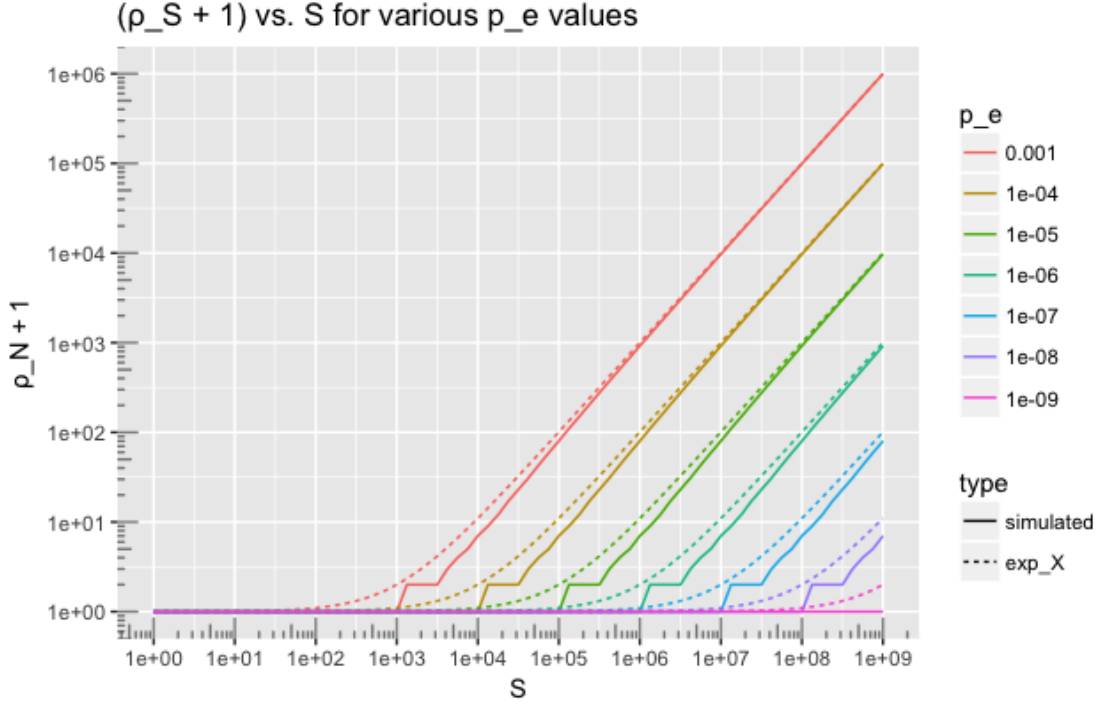


Figure 9: $(\rho_S + 1)$ vs. S for various values of p_e visualized using a $\log_{10}$ scale

- I only selected $p_e = 10^{-6}$ and $S = 10^7$ because they create a graph that illustrates both cases of q -values for low θ -values. This p_e value is easily possible for naturally-occurring morphemes in liaison-type systems, but 10^7 might not be a good value for S .

In this simulation (and running other simulations with a variety of parameters overwhelmingly yields the same result), $p_{S,\theta}^{(g)}$ falls to approximately $q_{S,\theta}$ within a relatively small number of generations — 20 seems to capture the overwhelming majority of cases — so we can estimate $q_{S,\theta} \approx p_{S,\theta}^{(40)}$ for good measure.¹⁷

We can see from this case that for all values of θ such that $q_{S,\theta} > 0$, it is also true that $q_{S,\theta} > 0.8$, and running a number of simulations with parameters set to a different values indicates that generally only high $q_{S,\theta}$ values and values of 0 are possible for reasonable values of S . This suggests that erosion of this sort is all-or-nothing; either total erosion occurs relatively quickly for a given morpheme or the morpheme's exceptional phonological behavior is retained by nearly all speakers with minimal erosion. This is more-or-less consistent with what we observe in French and English, with some caveats (which will be discussed in Section 3.4).

We also see that there exists a threshold which we will call ρ (in this case $\rho_{10^7} = 6$) such that $\forall \theta > \rho$, $q_{S,\theta} = 0$ and $\forall \theta \leq \rho$, $q_{S,\theta} > 0$. In other words, the exceptional phonological behavior of x is robust against erosion up to a θ value of at most $\rho = 6$. Formally, this robustness threshold is as follows:

$$\rho = \begin{cases} 0 & \text{if } \{\theta | \theta > 0, q_{S,\theta} > 0\} = \emptyset \\ \max_{\theta \in \{\theta | \theta > 0, q_{S,\theta} > 0\}} \theta & \text{otherwise} \end{cases}$$

We'll use the notation ρ_S to denote the value of ρ for a given estimate of S . Again, an upper bound on ρ_S can be easily ascertained using a statistical computing software package such as R using a simulation based on the upper bound of $p_{S,\theta}^{(g)}$.¹⁸ A visualization of $(\rho_S + 1)$ vs. S for various p_e values using a $\log_{10}$

¹⁷Even if there are cases where it takes a larger number of generations for equilibrium to occur, we're generally not interested in phonological changes that take more than 40 generations. If we are, we can simply increase the value of g we use

¹⁸For large ρ -values it is inefficient to check every θ value from 1 up to ρ , so a more efficient algorithm is necessary. The

scale¹⁹ is given in figure 9. From this figure, we can see that ρ_S as a function of S is roughly linear after $S = p_e^{-1}$ with an upper bound $\rho_S < p_e S = E[X_S^{(1)}]$, which makes sense — intuitively if the expected amount of disambiguating evidence is already less than the requisite threshold at the inception of the merger, we should expect erosion to occur. This also means we can use $\lfloor p_e S \rfloor$ as an upper bound on ρ without running a simulation (though the simulations run quickly so it is likely worth it to get the tighter bound from a simulation).

In principle we can generalize this approach to assess the robustness of any aspect of a language that requires the leveraging of alternations in determining grammatical behavior (e.g. irregular morphology, which relies on the irregular forms appearing in the stimulus).²⁰ However, we will not focus on such applications here.

3.1.2 Discussion of the Model

The model presented in the preceding section is an idealization and has a couple of limitations. Firstly, although we could reasonably estimate p_e for the surface realizations of a merging phoneme in any given morpheme based on data extractable from corpora, the model also depends on two less obvious parameters: θ and S .

How to estimate the true value of θ is not immediately obvious (e.g. it is unclear how the Tolerance Principle would apply, since here it is a question of learning exceptions to productive rules rather than of learning productive rules themselves). This means that we (as of yet) do not have a principled way of predicting a precise probability of whether phonological erosion should occur in a given word. It may be possible to use ρ to estimate a reasonably good value of θ , since the true value of ρ is a lower bound on θ for cases where erosion does occur and an upper bound on θ for cases where erosion does not occur (which we can observe). However, that θ is likely a function of some set of properties of the morpheme m in question, rather than a constant, and this set of properties and how they are weighted respectively might be difficult to determine, so this approach has its limitations. However, θ likely changes across languages (to be discussed in subsequent sections) and within languages at different points in time (alluded to in the preceding section), so this is an idealization.

Given accurate values of S and of θ , we would be able to predict that a given word will undergo erosion if $\rho < \theta$, even if our upper-bound estimate of ρ is loose (since the real value of ρ is less than θ by an even greater amount in this case). Additionally, since $\rho \geq 0$, one tautology we could easily identify is that erosion will occur if our upper-bound estimate of ρ is 0, since $\theta \geq 1$ and we know that $(\rho \leq 0 \wedge \rho \geq 0) \implies \rho = 0$, so $\rho < \theta$. Using this fact, we would be able to prove that erosion necessarily occurs for morphemes with sufficiently small p_e values. However, to reiterate, we do not know θ or S .

Even without knowing θ , we could still make some interesting claims given S based on the value ρ (which is determined only by p_e and S). Since we can only estimate an upper bound of ρ , it does not necessarily mean that x is robust against erosion if our estimate of ρ is greater than θ (since the true value could be less than both the estimate and θ), but we can see that a higher ρ value is correlated with a higher degree of robustness. Since our ρ -estimate is correlated with robustness against erosion, we can use it as a measure of how robust a given morpheme in a given language is against phonological erosion, even if we cannot always definitely predict whether or not erosion will occur in that morpheme. Hence, comparing ρ -values within and across languages could help explain differential behavior with respect to erosion following conditioned mergers of the sort discussed above.

However, again, we do not know S , and in some ways this parameter is even more problematic than θ .

algorithm I use finds a lower and upper bound on ρ in terms of $[10^k, 10^{k+1})$ and then repeatedly iterates through this interval with increasing granularity ($10^k, 10^{k-1}, \dots, 10^0$) to find tighter bounds until ρ is found, which is more efficient than checking every θ -value when $\rho > 10$.

¹⁹ $(\rho_S + 1)$ is used so that all $\rho = 0$ values are defined when displayed on a log scale

²⁰However, the usefulness of applying this model to other aspects of language, e.g. irregular morphology, is likely much more limited. Conditioned mergers are interesting (and possibly unique) in that they may introduce a large number of representational ambiguities simultaneously. French liaison, which in principle involves ambiguities in the phonological ending of every single morpheme, is a case in point. It is unclear what sort of morphological changes, for instance, could introduce ambiguity of the sort involved in the erosion model to such a massively broad array of cases

Firstly, it is known that the size of the linguistic input available to children is subject to significant variation, with children from more fortunate socio-economic backgrounds receiving more linguistic input (and correspondingly, developing a larger vocabulary earlier) (Hart & Risley 1995). If the “period of time during which a child is amenable to changes in their URs” on which S is based is connected to the end of the critical period, then this means that S is also subject to high variability across speakers, so the assumption that S is constants across time and speakers is again an idealization.

Although such assumptions about S are idealizations, they seem reasonable given that S indeed exists. However, a central way one might critique the model is to question whether S does indeed exist at all. It is true that there appears to be an end to the time window for learning productive rules (defined not by age but rather, by the number of environments N in which that rule could plausibly apply), as explained in the following quote (8) from Yang (2018):

- (8) ‘Furthermore, it is plausible to conjecture that acquisition never uses N ’s beyond a certain value, probably just a few hundred. It is doubtful that anyone can keep track of large values of N and e ; perhaps learners will simply freeze the rule once they have seen enough data, i.e. a sufficiently large value of N ’

However, we are not primarily concerned with the learning of productive rules, but rather with the learning of exceptions to productive rules. Particularly, if the closing of the window during which a productive rule can be learned is based on memory rather than developmental constraints, as (8) would suggest, then we do not have any reason to suppose that exceptions to productive rules cannot be learned even after the end of the primary L1 acquisition period. Adults who become fluent in an L2 are a case in point that productive rules as well as exceptions to them can be learned quite late. Within L1 scenarios, we also have reason to believe that exceptions can be learned late. One example related to the current work comes from French *h-aspiré*, which refers to vowel-initial morphemes that behave (exceptionally) as though they have an initial consonant, yielding strange behavior in relation to phonologically-conditioned allomorphy in preceding words, e.g. *le héros* /lə.ɛʁo/ */lə.ɛʁo/ ‘the hero’ with *h-aspiré* vs. *l’homme* /lɔ̃m/ */lə.ɔ̃m/ ‘the man’ without. It has been noted that loanwords, many of which are likely to be learned quite late, tend to have *h-aspiré*, indicating that an exception to the normal rule *surface vowel-initial* → *underlying vowel-initial* learned relatively late.

This contrasts a prediction of the model of the previous section, that there exists some value S such that any morpheme m learned after the size of the linguistic input has reached S will follow the regular rule R , i.e. m will not be learned as an exception to R . While we have the reasons given above to believe that this prediction is incorrect, I argue that we can still reasonably assume the existence of age during L1 acquisition and a corresponding value S such that *most*, even if not all of the words learned after S has been reached will follow regular, non-exceptional phonological behavior. It has been well-observed that older children and adolescents tend to adopt linguistic patterns resembling those of their peers (think of the sociolinguistic trope that sound change is innovated by middle school girls in the school yard imitating and incrementally overshooting the targets of their slightly older peers). Thus, it would appear reasonable to assume that individuals are less likely to adopt an exceptional UR for a given morpheme if the majority of their same-aged peers have also not adopted an exceptional UR for that same morpheme by the age where linguistic input from peers begins to outweigh linguistic input from parents as the individual develops their adult grammar and phonological representations. Even if the end of this window is not watertight, it therefore nevertheless seems reasonable to assume such a window in developing an idealized model of erosion of this sort.

Despite the aforementioned limitations and potential objections to the model, as an idealization it appears to perform quite well in terms of broadly illustrating the patterns of diachronic development of phonological systems dependent on disambiguating evidence for URs after conditioned mergers.

Firstly, it makes the correct prediction that the exceptional phonological behavior of morphemes implicated in the merger is either robust against or highly susceptible to erosion. This is consistent with evidence from the erosion of English /ɹ/-liaison and French liaison. The historic contrast between liaison consonants and $\emptyset$ morpheme-final was evidently unstable in English and for nearly all French morphemes, given the fact that exceptional liaison-related behavior (i.e. liaison itself in French and non-liaison in English) occurs for

only a small group of morphemes in French and apparently for none at all in English. The majority of cases where liaison still occurs in French generally have been around for centuries and do not appear to be going anywhere.²¹

Secondly, since the robustness of a given morpheme against erosion is directly a function of the value p_e , which is mainly determined by word frequency, the model predicts that word frequency plays a role in determining whether a morpheme is susceptible to erosion. Since the robustness measure ρ is upper bounded by Sp_e , which is a linear function of p_e , a core property of the model is that p_e values – and therefore frequency values – are directly correlated to robustness against erosion. This generally appears to hold up in French, where the set of morphemes that continue to exhibit liaison (i.e. exceptional phonological behavior) is generally restricted to highly frequent function words and inflectional endings, whereas less frequent morphemes (including all noun stems) no longer exhibit erosion outside certain set frozen expressions (and sometimes poetry).

In light of the generally correct predictions of the model, it seems a reasonable tool to use to explain and explore the different paths of the diachronic development of English /ɹ/-liaison and French liaison. However, since θ and S are not known, we will not attempt directly to utilize the model to predict which specific words should undergo erosion, but rather we will focus on the implication of the model that robustness against erosion is a function of word frequency and the degree to which observed general patterns of behavior are consistent with and divergent from the expectations of the model. Where behaviors are apparently divergent from expectation, we will attempt explain such scenarios on the basis of idiosyncratic features of the two systems and languages.

3.2 Background: English /ɹ/-Liaison

In the following sections, we will discuss relevant background regarding /ɹ/ liaison in non-rhotic English dialects, in particular with respect to the synchronic characteristics of the system in modern varieties, the genesis and early historic development of the system, and its acquisition by children.

3.2.1 Synchronic System of English /ɹ/-Liaison

English /ɹ/-liaison, also referred to as /ɹ/ sandhi, is frequently divided into two cases, linking and intrusive /ɹ/ respectively, on the basis of whether or not the /ɹ/ is etymological (and hence apparent in orthography), e.g. linking /ɹ/ in *the tuner is* vs. intrusive /ɹ/ in *the tuna is*, which may be pronounced homophonously as [ðə 'tjuːnəɪ ɪz]. The insertion of a liaison /ɹ/ may occur morpheme-finally before a vowel after any class of vowel phonemes that includes instances historically followed by /ɹ/. In modern RP English, this set is /ə, ɜː, ɑː, ɔː, ɪə, ɛə, ʊə/ (Hannisdal 2006, p. 106).

Even as linking /ɹ/ enjoys prestige in a number of dialects, including RP, intrusive /ɹ/ has been decried and stigmatized on prescriptivist, spelling-based grounds. This is apparently reflected in usage rates (which are variable for both /ɹ/-liaison types word-finally); Hannisdal (2006) finds much higher rate of linking than intrusive /ɹ/ (59.8% vs. 32.6% of possible environments) (p. 159, 169). However, this is generally considered strictly to be an effect of orthography rather than a difference of phonological representation, as illustrated by the following quote of (9) from Wells (1982, p. 224):

- (9) “In order to succeed in suppressing intrusive /ɹ/ while retaining linking /ɹ/, as a few do succeed in doing, the speaker must consult his knowledge of the spelling as the only guide to distinguishing the two cases.”

It is generally noted that linking /ɹ/ and intrusive /ɹ/ have similar distributional properties and Wells (1982) further notes that efforts to avoid use of intrusive /ɹ/ tend to result in an accompanying fall in rates of linking /ɹ/ as well (p. 224). The fact that Hannisdal found a lower rate of linking /ɹ/ than past studies might reflect

²¹Although there are some cases of liaison that are possible in more formal, likely unnaturalistic speech, where it does not occur in normal speech. One might argue that these cases are indicative of mild susceptibility to erosion, since the erosion is still in process centuries after the inception of the liaison system. This will be explored in following sections.

this observation, given that she analyzed speech from English newscasters who, being more speech-conscious, might have displayed this effect.

Intrusive /ɹ/ apparently occurs in all modern dialects areas in which linking /ɹ/ occurs, e.g. most of England, Australia, New York, New England, etc., and absence of intrusive /ɹ/ is associated with absence of linking /ɹ/, e.g. South African English, non-rhotic Southern American Varieties (Wells 1982). This further suggests a strong link between the two varieties of /ɹ/ liaison.

Analyses of /ɹ/-sandhi fall into two camps; those that assume underlying /ɹ/ with deletion accounting for surface forms in non-liaison contexts and those that assume no underlying coda /ɹ/ with insertion accounting for surface forms in liaison contexts. Although both analyses can be posited in a phonologically consistent way, the majority of past authors prefer the second analysis, since it is thought that an insertion analysis can more easily accommodate the existence of intrusive /ɹ/²² (e.g. Johansson 1973, Wells 1982, Brown 1988, Foulkes 1997). The exact nature of this rule is actually not relevant for our erosion analysis, but for sake of consistency with the majority of past literature, we will assume an insertion rule. Wells (1982) formulates this rule as follows in (10) (p. 226):

$$(10) \quad \emptyset \rightarrow \text{ɹ} / \left[\begin{array}{c} \text{V} \\ \text{-high} \end{array} \right] \text{-}\#_0 \text{V}$$

Such an analysis constitutes a rule inversion relative to the diachronic rule of /ɹ/-dropping that led to the genesis of /ɹ/-liaison systems, to which we turn in the next section.

3.2.2 Historic Origins and Development of English /ɹ/-Liaison

Clearly, a prerequisite for linking-/ɹ/ is morpheme-final /ɹ/ deletion. Relics of early deletion, particularly in pre-/s/ contexts, still exist in some English dialects, e.g. *ass* in American English and regional British dialects where standard British English uses *arse*, *cuss* for *curse* in regional varieties. However, systematic deletion of coda /ɹ/ did not start until the 16th and 17th centuries, and was not accepted as part of any prestige varieties until as late as the 1790s (Lass 1999, p. 114). On the development of /ɹ/-deletion in Southern British English (where it is considered to have originated), Lass offers the following (1999, p. 115):

- (11) “We can conclude that in less formal speech /r/-loss began sporadically in the fifteenth century; that in the seventeenth it had weakened post-vocalic allophones; and that in the later seventeenth and early eighteenth centuries it was generally still pronounced in all positions, but by the 1740s to 1770s was on the way to deletion, perhaps especially after low vowels. Weakening and loss became less variable and more codified as the century progressed; by the 1790s /r/-less pronunciations must have been very common, and increasing.”

It is generally assumed that /ɹ/-deletion could always be bled prevocally, implying that linking-/ɹ/ has been around since the beginnings of /ɹ/-deletion. With respect to the development of intrusive-/ɹ/, Wells notes the following (1982, p. 227):

- (12) “The earliest reference to intrusive /ɹ/ of which I am aware dates from 1762, when T. Sheridan mentions it as characteristic of London speech. It has probably characterized RP since the early nineteenth century, though no doubt regularly disapproved of and avoided by the speech-conscious.”

Based on Lass’s analysis, this would place the inception of intrusive-/ɹ/ right around the time that /ɹ/-deletion became invariable, suggesting it was a relatively immediate result of invariable /ɹ/-dropping in a bleeding order with resyllabification with a following vowel. This is consistent with the prediction of the model that erosion should occur quickly after inception of the merger.

²²However, deletion analyses are not necessarily inconsistent with intrusive /ɹ/; For example, depending on the frameworks we use, children could plausibly learn a phonotactic constraint against vowels involved in liaison from occurring morpheme-finally, causing them to posit underlying /ɹ/ in both linking and intrusive /ɹ/ words that just happens to undergo deletion most of the time

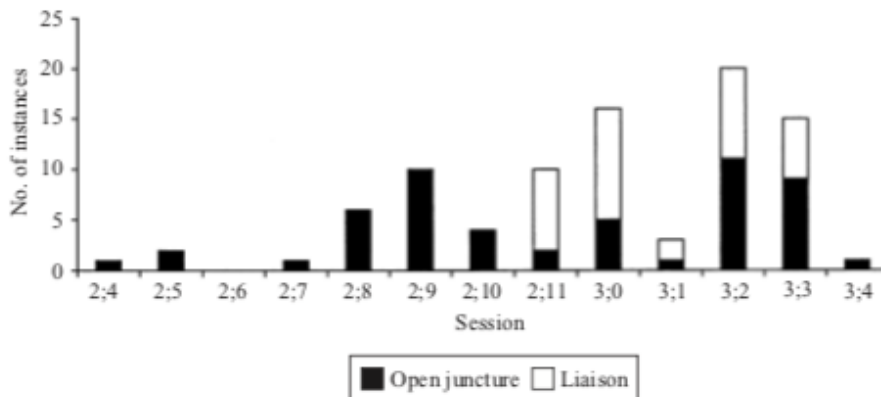


Figure 10: Realizations of /ɹ/ liaison environments (Newton & Wells 2002)

3.2.3 Acquisition of English /ɹ/-Liaison

Newton & Wells (2002) found in a longitudinal study of a child from North London a complete lack of all types of /ɹ/ liaison up to age 2;10, with a sudden leap in rates of liaison at age 2;11, as shown in Figure 10. The authors do not note any differences in observed rates of linking and intrusive /ɹ/, consistent with the analysis of the two phenomena as phonologically identical. In general, there appears to be much less complexity involved in the acquisition of /ɹ/-liaison than in French liaison, which we will discuss in Section 3.3.3.

3.3 Background: French Liaison

In the following sections and in parallel with Section 3.2, we will discuss relevant background regarding French liaison, in particular with respect to the synchronic characteristics of the system in modern French, the genesis and subsequent historic development of the system, and its acquisition by children. In general, we will see that the situation is much more complex for the French liaison system than for the English /ɹ/-liaison system described in Section 3.2 above.

3.3.1 Synchronic System of French Liaison

Similar to /ɹ/-liaison in English, French liaison involves the appearance of otherwise absent consonants word-finally before a vowel-initial word, e.g. *les* ‘the (pl.)’ pronounced with a final [z] prevocally in phrases such as *les hommes* [lezɔ̃m] ‘the men’ but with no such final consonants in isolation or preconsonantly as in *les femmes* [lefam] ‘the women’. Different from the English case, however, French liaison involves multiple possible liaison consonants, namely /z, t, n, ʁ, p, g/ (Fagyal et al. 2006, p. 67).

However, the overwhelming majority of liaison instances occur with only three of these consonants: /z, t, n/ (Posner 1997, p. 264). Liaison with /ʁ/ occurs frequently only with a handful of adjectives that precede nouns, namely *léger* ‘light’, *premier* ‘first’, and *dernier* ‘last’, though it may occur rarely in a “very formal style, possibly pedantic or even pompous” with first conjugation infinitives in *-er*, e.g. *aimer un peu* [ɛmɛʁpø] ‘to love a little’, normally pronounced [ɛmɛʁpø] (Tranel 1987, p. 174-175). Liaison with /g/ only occurs in the adjective *long* ‘long’ when it precedes a noun, e.g. *un long été* [ɛ̃lɔ̃gete] ‘a long summer’. The historic liaison consonant for orthographic /g/ was actually /k/ (due to historic word-final devoicing), though this pronunciation is no longer found except in the frozen expression *sang impur* [sãkɛ̃pyʁ] ‘impure blood’ known from the French national anthem and is otherwise perceived as archaic (Tranel 1987, p. 174). The occurrence of a liaison /g/ in *long* likely derives from analogy with the feminine form *longue* /lɔ̃g(ə)/.

Finally, liaison with /p/ is restricted to only two adverbs: *trop* ‘too much’ and *beaucoup* ‘many’ (Tranel 1987, p. 174).

Even among the three common liaison consonants, it has been noted that even these occur with a relatively low frequency and that these cases often serve a morphological purpose, “e.g. [z] as a plural marker: [le.za^hbʁ] *les arbres*, [i.zaʃɛt] *ils achètent*; [n] mainly with grammatical elements like *un*, *en*; and [t] as a verb inflection, as in *c’est-à-dire*, *vient-il?*” (Posner 1997, p. 264).

One peripheral phenomenon in some ways analogous to English intrusive /ɹ/ is the phenomenon of false liaisons or *pataquès*, of which there are two main types: *velours* (‘velvet’) involving /z/ and *cuir* (‘leather’) involving /t/ (Tranel 1987, p. 170). Velours occur particularly often with modifiers indicating plurality, such as numerals, likely a result of the function of liaison /z/ as a plural marker; “*vingt* [vɛ̃] ‘twenty’ and *cent* [sɑ̃] ‘one hundred’, which theoretically link with [t] [...] are often heard entering into liaison with [z]” (Tranel 1987, p. 179). However, such instances of velours remain “outside the norm” (Tranel 1987, p.170).²³ This, combined with the general perception of pataquès as erroneous (rather than stigmatized as bad style), likely suggests a hypercorrection phenomenon rather than a phonologically regular instance of rule inversion as in the case of intrusive /ɹ/ in English. However, limited cases where historic false liaisons have been incorporated into the standard grammar of the language do occur and will be discussed in the next section, Section 3.3.2

A large number of generalization can be made about the cases in which liaison may and may not occur. Liaison is generally considered obligatory between a determiner and a following noun or adjective, between an adjective and a following noun it modifies, between certain prepositions (*chez*, *sans*, *dans*, *en*, etc.) and between a verb and adjacent pronominal clitics, and between two adjacent pronominal clitics themselves. It is never permitted after a singular noun outside certain frozen expression, though it is permitted after a plural noun and a following adjective in elevated styles. Some syntactic constraints are also observed, e.g. a noun phrase subject never participates in liaison with a following vowel-initial verb. Most remaining contexts allow liaison optionally depending on register, with varying levels of perceived formality. (Tranel 1987, p. 183-190)

Based on these facts (which are only a simplification), a variety of phonological analyses have been offered. A common analysis using non-linear phonology involves positing that liaison consonants are underlyingly ‘floating’ unsyllabified consonants, that are deleted by stray erasure unless syllabified with a following vowel-initial word (Posner 1997, p. 264). Other approaches involve more morpho-phonological analyses. In particular, liaison with adjectives has been analyzed as phonologically motivated suppletion between vowel-final masculine forms and consonant-final forms that are usually feminine (Tranel 1996, Steriade 1999). We will revisit suppletion-based explanations of liaison in adjectives in Section 3.5.2, but otherwise, the exact phonological mechanics of liaison are generally not of major relevance in our analysis of liaison; regardless of the exact mechanics, a child in acquisition will still need to leverage disambiguating evidence in determining in which words liaison may occur and with which consonants such liaisons take place, so we still should expect erosion as predicted by our model regardless of the exact phonological analysis.

3.3.2 Historic Development of French Liaison

Again as with English /ɹ/-liaison, French liaison developed from the historic deletion of word-final consonants, which could be bled by resyllabification with a following vowel-initial word (called *enchaînement* in the context of French liaison). Tranel summarize the gradual deletion of word-final consonants as follows (1987, p. 169):

- (13) “In Old French, final consonants were pronounced, but from the twelfth to the sixteenth centuries, they progressively disappeared, first in preconsonantal position and then at the pause, leaving them to appear only in prevocalic position. Later, other restrictions came to reduce even more the contexts in which these consonants could appear, so much so that liaison today occurs far less than it used to (but probably more than tomorrow).”

²³Though Tranel (1987) notes the exception of the expression *entre quat’yeux* [ɑ̃tʁœkatʃjø] ‘between you and me’ as a fixed phrase with a numeral-related velours (p. 170)

This word-final consonant deletion was complete by the sixteenth or seventeenth century, and occurred generally for all word-final consonants except /l/, consistently retained, and /ʁ/, frequently deleted though sporadically retained (Posner 1997, p. 263-264). Posner posits that final liaison consonants were still part of the underlying representations of words (1997, p. 264). At early stages after final consonant deletion, such an analysis could be postulated without relying on non-linear phonology with floating segments or on other particularly abstract frameworks, since the absence of word-final consonants outside liaison contexts meant a simple deletion rule, bleedable by *enchaînement*, could accommodate all the facts.

However, two changes soon complicated the phonological system, rendering such simple analyses untenable. Firstly, right around this time word-final /ə/, which had protected consonants from deletion, particularly in feminine words, started undergoing deletion, producing a new set of words with word-final consonants, e.g. *bonne* [bɔ̃n] ‘good (f.)’ vs. *bon* [bɔ̃] ‘good (m.)’. Although this was not wholly incompatible with positing simple word-final consonants in URs with a deletion rule — we could simply postulate that /ə/ remained underlying with a /ə/-deletion rule that counterfeeds word-final consonant deletion — a later change involving the restoration of word-final consonants in some words (usually considered instances of spelling pronunciation), e.g. *cinq* ‘five’, *porc* ‘pig’, *tabac* ‘tobacco’, *dot* ‘dowry’, etc., rendered word final consonant deletion unpredictable without adding some further level of abstraction, e.g. in the form of floating liaison consonant segments (Posner 1997, p. 291).²⁴

Another change worthy of note is the incorporation into the standard language of certain overgeneralizations of liaison. In particular, a liaison /t/ has become obligatory between a third person verb and an inverted subject starting in the 16th century. This liaison is a regular expected result of liaison for a large number of verbs, e.g. *vend-il?* [vɑ̃til] ‘does he sell?’, *finit-elle?* [finitɛl] ‘does she finish?’, though it even became obligatory for third person singular present verbs of the first conjugation (verbs in *-er*), where no final /t/ had existed for about four centuries, e.g. *aime-t-il?* [ɑ̃mtil] ‘does he love?’ (Tranel 1987, p. 170). Similar to the unetymological /t/ in the third person present singular indicative, this period also gave rise to non-etymological /z/ in second person singular imperatives of first conjugation verbs when followed by the pronominal clitics *en* or *y*, e.g. *manges-en* [mɑ̃ʒzɑ̃] ‘eat some’ (with *s* incorporated into the orthography) vs. regular *mange un peu* [mɑ̃ʒɛpø] ‘eat a little’ (Tranel 1987, p. 170-171) That these changes developed around the time that final consonant deletion became categorical suggests that it was somehow an immediate result of the bleedable consonant-deletion rule (similar to the case with intrusive /ɹ/ and /ɹ/-vocalization).

3.3.3 Acquisition of French Liaison

A substantial literature has developed around the acquisition of French liaison. During early acquisition, a high rate of liaison-related learning errors are reported, particularly in noun phrases (which makes sense, as noun phrases frequently include obligatory liaison environments with a preceding article or adjective. Apparently a large proportion of these errors are tied to misparsing word boundaries, e.g. a liaison consonant is incorrectly analyzed as the initial consonant of the noun: *un ami* [ɛ̃nami] ‘a friend (m.)’ parsed as /ɛ̃ + nami/, leading to errors such as *les amis* *[lɛnami] ‘the friends’ where [lezami] is required, or an initial consonant is incorrectly analyzed as a liaison consonant: *un nuage* [ɛ̃nɥaʒ] ‘a cloud’ parsed as /ɛ̃n + ɥaʒ/, yielding errors like *les nuages* *[lɛzɥaʒ] ‘the clouds’ where [lenɥaʒ] is required (Buerkin-Pontrelli et al. 2017).

Buerkin-Pontrelli et al. (2017) discuss two competing analyses of the acquisition of liaison, namely a constructionist model (supported by Dugua (2006), Chevrot et al. (2009)) and a phonological model (Wauquier-Gravelines & Braud 2005). Both models hold that children initially represent noun phrases as unanalyzed chunks during Stage 1. In the constructionist model, children enter Stage 2 around age 3;0, at which point use schemas that parse noun phrases as determiners + /X/, with liaison consonants assumed to be part of the UR of the noun /X/, resulting in the sorts of errors described in the preceding paragraph. Finally, around age 5;0, children attain target pronunciations through the employment of schemas that parse noun

²⁴Curiously, word final consonants of this type without orthographically indicated /ə/ still allow [ə] in surface forms, e.g. *lac* /lak/ [lakə], in a case of ‘intrusive’ /ə/ with parallels to intrusive /ɹ/. While this is also a case of erosion associated with the variable deletion (i.e. merger with Ø) of /ə/, with its whole own set of peculiarities, this phenomenon is outside the scope of the current work

phrases as *det + /LCX/* (where LC is the liaison consonant required by the determiner). (Buerkin-Pontrelli et al. 2017)

In contrast to the constructionist model, the phonological model of (Wauquier-Gravelines & Braud 2005) posits that enter stage 2 somewhat earlier (aged approximately 2;0), and that their errors are motivated not by constructionist schemas, but by syllabic templates strongly preferring CV syllables. This preference may result in errors of the type described in the first paragraph of this section, but also in other types of errors not obviously related to the liaison consonant, including reduplication (*les amis* *[le.ma.mi]), idiosyncratic onsets (*les amis* *[le.ba.mi]), and truncation (*les amis* *[le.mi]), which are not predicted by the constructionist model. Finally, at age 3;0 (considerably earlier than the constructionist model posits), children attain adult-like liaison production. (Buerkin-Pontrelli et al. 2017)

Buerkin-Pontrelli et al. (2017) find errors of the type predicted by the phonological model and relatively early mastery of liaison within noun phrases, noting “our findings suggest that by age three children are well on their way to adult-like representation of liaison” (p. 215).

Although these facts indicate that French-speaking children are able to acquire liaison relatively quickly, it also highlights inherent difficulties in correctly identifying liaison consonants. In particular, in order for an instance of liaison to be correctly identified, the child must have knowledge of both words involved in the liaison. For instance, in order to know that *les* has a liaison /z/, not only must the child hear instances of *les* in a liaison context, but the child must also know that the vowel-initial word following *les* is indeed vowel-initial and not /z/-initial. Given the high frequency of determiners like *les*, we can expect this to occur relatively quickly (accounting for the early acquisition of liaison in such contexts), though one imagines this task to be much more difficult for less frequent words (which are already more susceptible to erosion given the assumptions of our mathematical model).

3.4 Direction of Erosion

As discussed earlier, the general directions in which the erosion of the two liaison systems dealt with in this work have trended are opposite: overgeneralization in English vs. loss in French. In the following subsections, we will attempt to explain this difference on the basis of the principle of Invariant Transparency and the Tolerance Principle, previously given in (1) and (4) respectively. In the following sections, frequency data is extracted from the SUBTLEX corpus (a corpus of 51 million words from American English subtitles) for English and the frTenTen12 corpus (a corpus based on 11.4 billion words taken from a crawl of the French internet, including lemma-based frequencies and parts of speech).

3.4.1 Overgeneralization of English /ɹ/-Liaison

The fact that we observe overgeneralization of English /ɹ/-Liaison from around the time that /ɹ/-dropping became categorical strongly suggests that the linguistic stimulus is such that children are motivated to posit a productive /ɹ/-insertion rule. To reiterate the Tolerance Principle (Yang 2016) as given in (4), a rule R is expected to be productive if the number of exceptions is less than θ , where $\theta := N/\ln(N)$ and N is the number of environments in which R could apply. In this case, we want to show that the number of words without etymological /ɹ/ are less than θ . To calculate this, I took the Penn Language Variation and Cognition Lab’s copy of the CMU pronouncing dictionary, extracted all linking and intrusive /ɹ/ words, and ordered the words by frequency.

In looking at productivity of intrusive /ɹ/, we should only consider contexts after the subset of /ɹ/-liaison phonemes /ə, ɔ:, ɑ:/ (the vowels after which intrusive /ɹ/ occurs). Figure 11 shows that productivity indeed holds for all N up to a 1000 (except for briefly at $N = 40$), assuming that words are acquired in order of frequency.²⁵ This makes sense given well-known facts in particularly regarding word-final occurrence /ə, ɑ:/ (the only vowels that occur with intrusive /ɹ/ with any significant frequency). The phone /ə/ only occurs word finally without historic /ɹ/ in non-Germanic words (c.f. Wells (1982): “Middle English had no native

²⁵Above about 1200, productivity no longer holds, but recall (8), which states that we only need to take into consideration N -values up to a few hundred

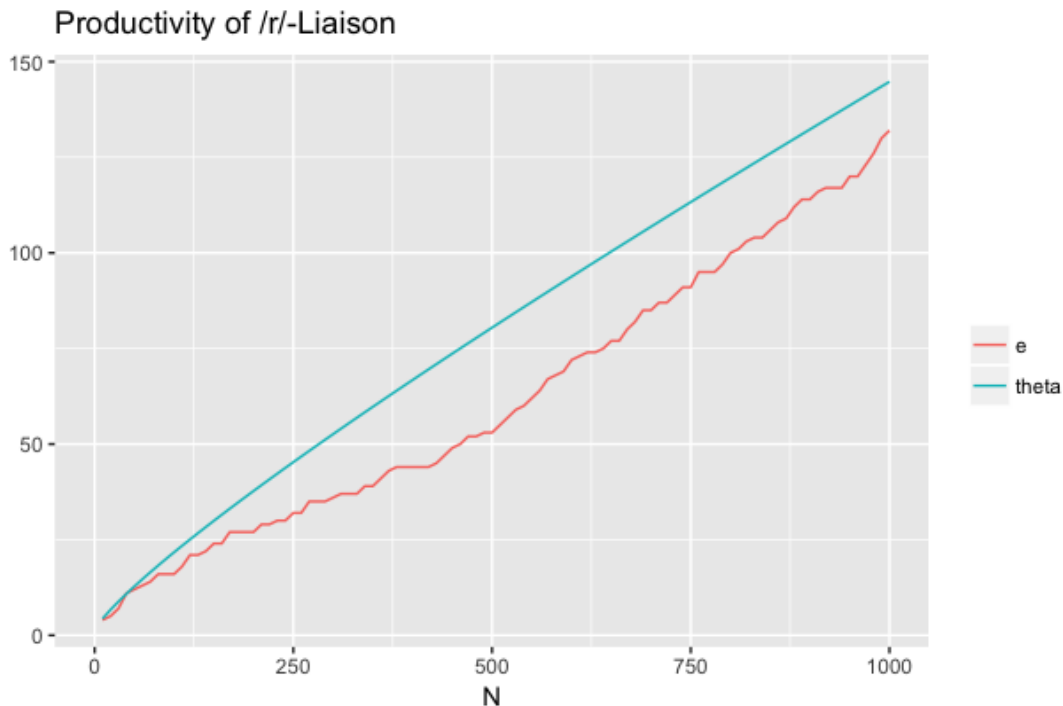


Figure 11: Productivity of /ɹ/-liaison based on *e* and *θ* vs. *N*

words ending in /ə/. All *comma* words... are borrowings from other languages, many of them belonging to specialist or learned vocabulary.” (p. 221)).

However, /ɔ/ does occur with reasonable frequency in some native English words, e.g. *saw*, *law*, *paw*, etc. To check that this phoneme class did not show a difference in productivity, we can look specifically at that vowel class, shown in Figure 12. Here we see that /ɹ/-insertion is no longer expected to be productive after /ɔ/ when *N* is in the mid-60s. However, this is not necessarily problematic for the productivity of /ɹ/-insertion for two reasons. Firstly, there are only 98 words types in the sample, so by the mid-60s we start encountering less frequent words like *macaw* and *guffaw*, which are likely not to be learned until later. Secondly, /ɹ/-liaison is variable even for etymological linking /ɹ/. This means that instances of non-liaison cannot necessarily be taken as evidence by the child that that word is an exception to /ɹ/-insertion; having observed the variability of the rule, they might simply assume that this was an arbitrary case of non-application rather than a systematic reflection of that morpheme’s underlying phonological behavior. Hence, we can reasonably assume the the inferred number of exceptions is deflated compared to the number we would estimate if we consider each originally /ɹ/-less word an exception.

We do not need to worry about accounting for other vowel phonemes, e.g. /ɪə, ɛə, ʊə, ɜ:/, since they generally all have historic /ɹ/. Hence, we have shown that /ɹ/-insertion is productive, which explains the direction of the erosion of the historic /ɹ/ - ∅ contrast in favor of /ɹ/ by overapplication of /ɹ/-insertion.

3.4.2 Loss (and Restricted Overgeneralization) of French Liaison

In the French case, where we have loss of liaison, we instead want to show that no productive liaison rule can be learned (with the exception of certain subcases to which we will turn shortly). Intuitively, it makes sense that no productive liaison-consonant could be identified given the fact that at least 6 liaison consonants are involved in the system (and even more historically). For good measure, we can investigate the productivity of French liaison for the three most common liaison consonants again using the Tolerance Principle. To do this, I downloaded a lemmatized list of French words and grouped them based on final consonant (lack thereof

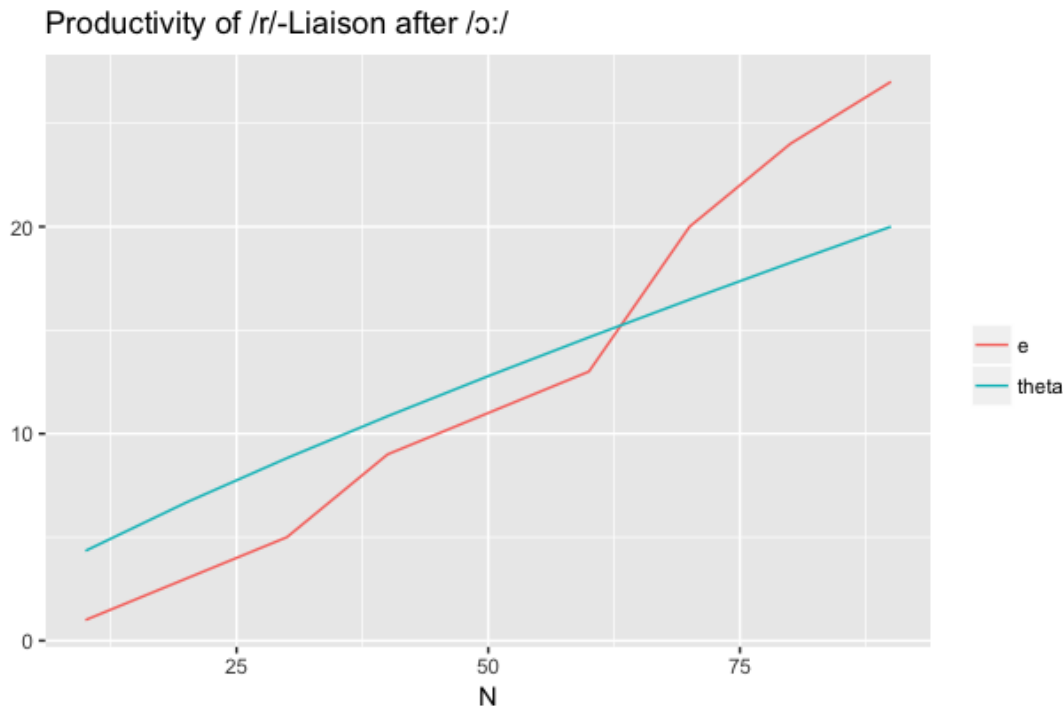


Figure 12: Productivity of /r/-liaison after [ɔ:] based on e and θ vs. N

for historically vowel-final words). The reasoning for using lemmatized words are as follows: inflectional endings like the plural /z/ as well as both /z/ and /t/ for verb inflections should likely be treated as single morphemes in our count of N , so we do not want to inflate the number of instances of /z/, for instance, by including every single plural word, when really these all collectively only represent one word-final morpheme. I filtered out verbs (which we can assume always end in some sort of inflectional morpheme) for this same reason. The end result is the graph of Figure 13, where each of the three dominant liaison consonants fails to meet productivity even at low values of N . Given that no productive rule can be found (in most cases), we expect the principle of Invariant Transparency (1) to determine the default behavior of word-final $\emptyset$, namely with the rule *surface word-final $\emptyset$ = underlying word-final $\emptyset$* .

However, we observe that rule inversion did indeed occur resulted in a restricted case of overgeneralization leading to non-etymological liaison instances for third-person singular present /t/ and second person singular imperative /z/ in first conjugation verbs with infinitives in *-er*. To test productivity, I took a list of verb lemma frequencies based on frequency and marked 1st conjugation verbs as exceptions.²⁶ Using the simplifying assumption that verbs of the two types (with or without the etymological liaison consonant) occur with the same relative frequencies across the entire lemma and within the specific inflected forms in question (i.e. 3rd person singular present and 2nd person singular imperative), we get the graph of Figure 14 when we apply the Tolerance Principle.

Figure 14 shows that the /t/ and /z/ insertion rules are productive until about 26 verbs are learned, but then should be expected to become non-productive. There are a few possible explanations for why productivity holds (evidently) in spite of this finding. Firstly, we might consider the possibility that liaison in these two cases might have been variable historically. If this were the case, then we again might be able to attribute unexpected productivity to the fact that non-application of liaison could not necessarily be taken by the child in acquisition as definitive evidence of liaison as impermissible in those cases. Secondly (and likely more

²⁶Verb lemmas are easier to use than non-lemmatized frequencies in this case because of syncretism between the 1st and 3rd person singular present and 2nd person singular in the 1st conjugation, which would inflate the apparent number of exceptions to productive /t/ and /z/ rules if we were to simply include verb forms that *could* match the structural description of the /t/ and /z/ rules based on spelling.

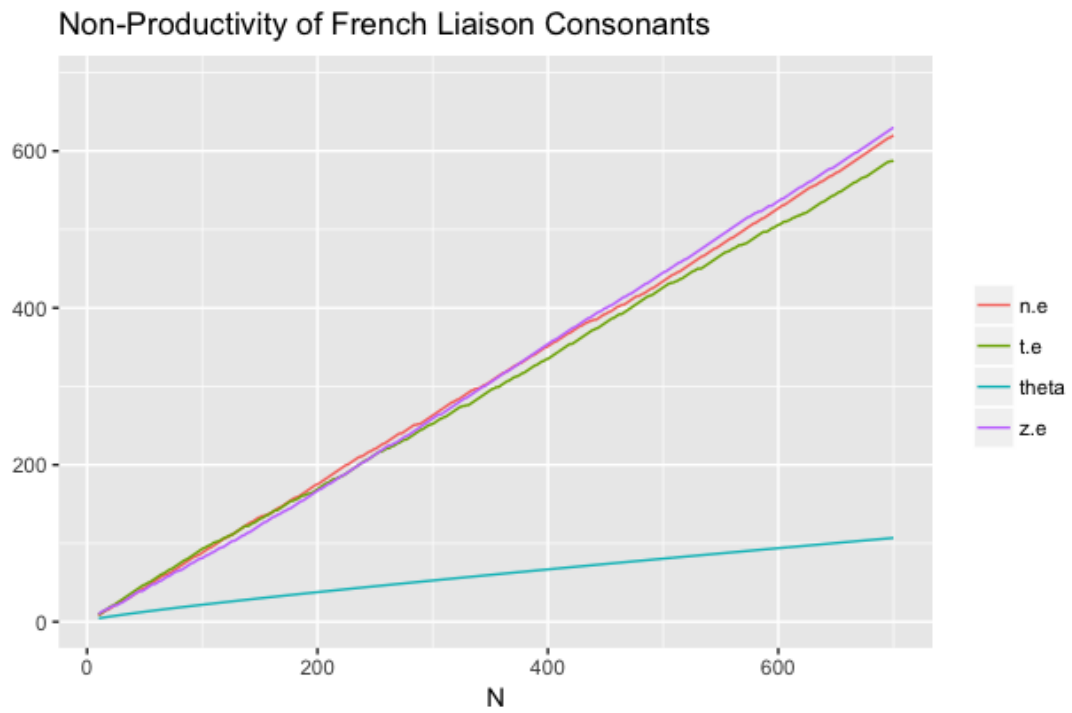


Figure 13: Nonproductivity of Top Three Liaison Consonants (exceptions to default /n/, /t/, and /z/ rule represented n.e, t.e, and z.e respectively)

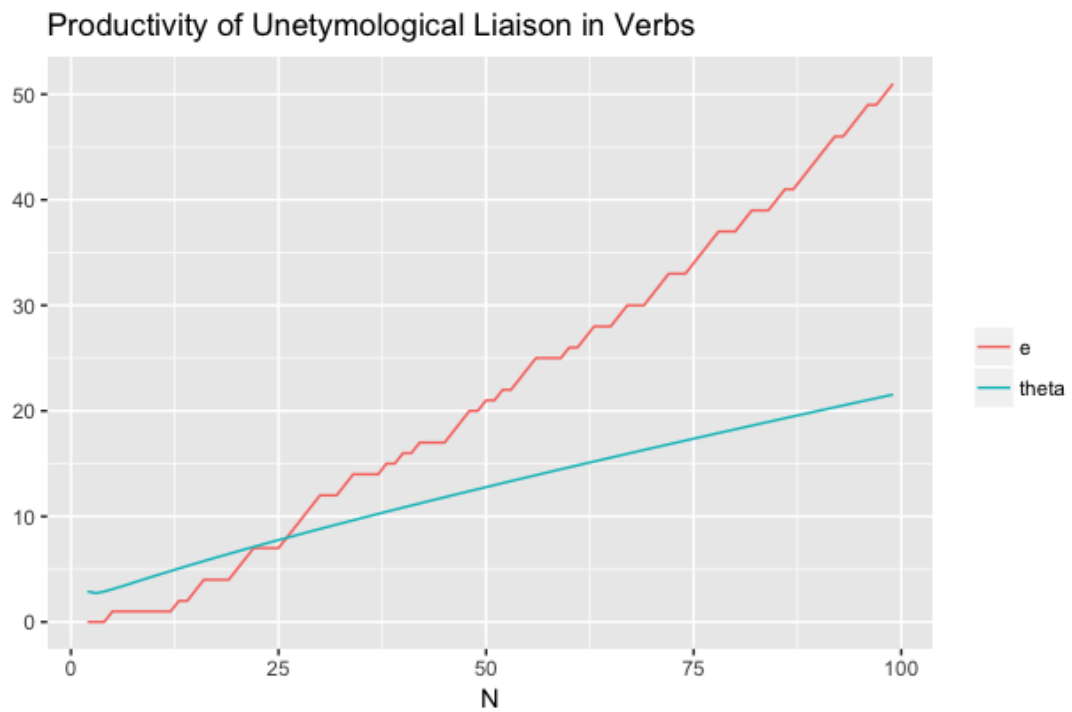


Figure 14: Productivity of Nonetymological Liaison in 1st Conjugation Verbs

critically), /t/ and /z/ mark 3rd and 2nd person respectively in verb forms other than the two where the grammaticalized overapplication of liaison occurs. Then, the fact that /t/ marks third person plural for all verb forms and third person singular for 3rd person singular imperfect and conditional forms, for example, might facilitate the acquisition of overapplied /t/-liaison. This is especially likely to hold true if we were to investigate the productivity of /t/ liaison as a marker of third person more generally, using a broader set of environments for our definition of *N* when attempting to apply the Tolerance Principle. Analogous (and perhaps stronger) claims can be made about /z/ and the second person, which is always etymologically present in the 2nd person plural and in all tense forms in the 2nd person singular apart from the imperative.

3.5 Degree of Erosion

We still have to account for differences in degree of erosion. To reiterate, erosion of the historic contrast between liaison consonants and $\emptyset$ is complete in the English (i.e. intrusive /ɹ/ can occur in any word). Even if the direction of erosion in both the English and the French case has been explained, it is not immediately clear why erosion in English should be so much more extensive than in French. Unambiguous alternants indicating participation and non-participation in liaison (which yield disambiguating evidence of phonological behavior), are expected to have historically occurred in approximately the same set of environments, i.e. pre-vocally across word boundaries. We can estimate the probability that a following word is vowel initial by summing the probabilities of a representative set of vowel initial words occurring (which can be taken from frequency). This gives us a rate of about 24% of English tokens with an initial vowel and 29% of French tokens. Hence, given that the disambiguating environment occurs at a roughly equal frequency for both cases, it is not immediately clear why erosion should have been so much more complete for English than French. The next two subsections investigate this problem.

3.5.1 Total Erosion in English /ɹ/-Liaison

One obvious answer to why total erosion has occurred in English is the set of English morphemes without historic /ɹ/ (which in principle could have retained an exceptional lack of intrusive /ɹ/ after the development of /ɹ/-liaison) are simply less frequent than words with analogous exception behavior (i.e. liaison words) in French. French liaison largely occurs in function words and inflectional endings with generally high frequencies, whereas the most common intrusive-/ɹ/ word that regularly participates in /ɹ/-liaison in English is *saw*, which – although quite frequent – occurs nowhere near as frequently as the ubiquitous French plural morpheme /z/, for instance.

However, there is some reason to doubt this interpretation. Notably, a number of non-rhotic varieties with reduced variants of highly frequent function words like *you* or *to* ending in /ə/ allow /ɹ/-liaison, e.g. *to eat* [təɪ 'i:t] in some British Englishes (Wells 1987, p. 227), even though we should expect such words to be protected from erosion due to their frequency if the above interpretation holds. Another explanation might be the fact, already mentioned, that the variability of /ɹ/-liaison limits the usability of /ɹ/-less realization as evidence that /ɹ/-insertion is disallowed, which, turning back to our mathematical model, has the effect that the child in acquisition will need a much higher threshold θ of disambiguating evidence for erosion to be avoided. Yet another explanation may be that /ɹ/-insertion is a productive rule, which a lack of /ɹ/-insertion requires to be suppressed. In contrast, the Principle of Invariant Transparency should likely not be viewed so much as a productive rule in a grammatically active sense (once speakers assign a word a transparent underlying form, they do not need to perform any further operations on that UR to get surface forms, in contrast to rules like liaison or /ɹ/-insertion), but rather as a heuristic for assigning underlying representations. Thus, liaison might not require suppression of productive rules in the same way that a lack of /ɹ/-insertion does.

3.5.2 Partial Erosion in French Liaison

As already discussed, non-erosion (i.e. occurrence of liaison) in French is by-and-large associated with function words and inflectional endings, whose high frequency, as predicted by our model, results in robustness

against erosion. The fact that liaison has totally eroded in noun-stems (i.e. singular noun forms) is also in line with the frequency-based explanation which we used to explain erosion in English and which is predicted by our model.²⁷

One potential hiccup with this interpretation is the existence of a number of comparatively lower-frequency adjectives in French that still have liaison. However, these are all masculine adjectives, most of whose liaison forms are identical with their feminine equivalents, so it may be possible to explain this through productive phonologically-conditioned syncretism, meaning that the liaison forms do not need to be learned in the liaison contexts, which would make such forms more robust against erosion in spite of their comparatively lower frequency. The fact that these liaison forms are likely based on feminine forms rather than from direct observation of the masculine forms in the liaison context receives support from the fact that some modern pronunciation standardly differ from their historic form (e.g. *long* has the liaison form /lɔ̃g/ ‘long (m.)’ resembling the feminine form *longue*, even though the historic form was /lɔ̃k/; adjectives like *ancien* /ɑ̃sjɛ̃/~/ɑ̃sjɛn/ ‘old’ lose their nasalization in liaison forms, even though other morphemes with /n/-liaison do not show this behavior, e.g. *un* /ɛ̃(n)/ ‘a (m.)’). Additionally, a suppletion rule is likely needed to explain certain particularly irregular masculine adjectives anyway, including *vieu* [vjø] ‘old’ and *beau* [bo] ‘beautiful’ with pre-vocalic forms *vieil* [vjɛj] and *bel* [bɛl] that are phonetically identical to feminine forms *vieille* and *belle*. Indeed, Tranel (1987) notes that “Concerning singular masculine adjectives, their linking consonants are phonetically related to the endings found in the corresponding feminine forms” (p. 179). We could likely use the Tolerance Principle to show the productivity of syncretism of this sort, though that remains outside the scope of this work.

3.6 Section Summary

Whereas Section 2 was concerned with lack of disambiguating evidence for phonological representation after conditioned merger, in the current section (Section 3), we have concerned ourselves with phonological systems that involve conditionally merged phones whose phonological representation and behavior may regularly be disambiguated through environmental alternations. Nevertheless, even given the presence of disambiguating evidence, difficulties in learning phonological representations arise, resulting in at times curious patterns of phonological change, in particular the erosion of the historical contrast between the two merging phonemes by changes favoring one potential representation over the other.

In subsection 3.1, we showed that such erosion may be an intrinsic result of the limitations of the disambiguating evidence available, and developed a mathematical model that makes certain predictions about the nature of such erosion processes. Firstly, we predict that more frequent morphemes with alternants in and outside the environment of the conditioned merger are more robust against erosion. Secondly, we predicted that robustness against erosion is binary for most cases, and that words should either be robust against erosion (and not undergo such changes) or be highly susceptible to erosion, in which case erosion is expected to occur relatively quickly.

In the subsequent subsections, we considered the erosion of two liaison systems, namely English /ɹ/-liaison and French liaison. Although the genesis of each system bears great resemblance to the other as the result of word-final consonant deletion that could be bled by resyllabification with a following syllable, the two have followed markedly distinct paths of development. Specifically, the English case has demonstrated overgeneralization of liaison in the form of unetymological intrusive /ɹ/, while the French case has generally demonstrated loss of liaison consonants, morphemes, and contexts, with the exception of a couple of early instances of overgeneralization.

Although the differential development of the two systems in question are striking, in a certain sense these differences are superficial, as these developments still represent erosion of a historic phonological contrast in the direction of a more easily applicable default phonological rules. In Section 3.4, we attempted to explain the direction of erosion in both cases on the basis of the Tolerance Principle, which showed liaison to be productive in the English /ɹ/ case, but not in the French case.

²⁷Note, some infrequent liaisons in formal speech that seem incongruent with the prediction of erosion in such infrequent cases are likely an effect of prescriptive linguistic conservatism, and do not reflect the regular expected path of development, seen more clearly in informal, colloquial usage

In Section 3.5, we further discussed peculiarities of each system, particular with respect to the extent of erosion in each case. We found that a larger number of instances of non-erosion in the French case is likely a combined result of a larger number of affected morphemes with a high frequency (predicted by the model of 3.1 to be robust against erosion) as well as difficulties in learning exceptions to /ɹ/-liaison in English associated with the variability of the phenomenon, in addition to some peculiar aspects of French involving syncretism of adjectives.

4 Conclusions

In this thesis, we focused on the effects of conditioned mergers on underlying representations, resulting from ambiguities that such mergers inevitably introduced to the system. This ambiguity of phonological representation presents a formidable task to the child in acquisition, who now faces the challenge of deciding on an appropriate underlying representation on the basis of at best limited linguistic evidence.

In some cases, as in the case of intervocalic flapping in Section 2, there may be no direct evidence in the stimulus for how this ambiguity should be resolved, which presents a problem not only to the child learning the language, but also to us as linguists in trying to understand how individuals represent phones learned under such conditions. The current work found some evidence that orthography plays a role in the phonological representation of ambiguous phones (or at the very least in their production), though the precise extent and nature of the relationship between orthographic and phonological knowledge remains uncertain and is left open to future work.

Even in scenarios where disambiguating evidence is available, we observe peculiar patterns of diachronic phonological development, as exemplified by our case study of two liaison systems in English and French. Apparently, such developments are associated with inherent limitations on the child’s ability to leverage such disambiguating evidence in developing phonological representations for ambiguous surface phenomena resulting from conditioned mergers. We considered a mathematical model that strongly predicts the occurrence of such erosion when disambiguating evidence occurs for only a subset of tokens of a phonologically ambiguous morpheme, with the exception of particularly frequent morphemes that are robust against erosion of this sort. Erosion of phonological systems resulting from conditioned merger evidently may follow a variety of paths, differing both in direction (having to do with the plausibility of potential productive rules) and extent of the erosion (having to do with individual word-frequency and various language-specific factors).

More broadly, the current work indicates that conditioned mergers and the associated ambiguity that they introduce may have drastic effects on phonological representation, resulting in synchronic phonological systems that are simultaneously fascinating and puzzling.

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